A Mean-Field Game of Mayoral Endorsements: Identification and Evidence from Brazilian Coalitional Presidentialism

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A mayor's return from endorsing a presidential candidate falls as other mayors endorse the same one. Endorsement is thus strategic, and it has gone structurally unestimated because a discrete-choice game among Brazil's 5,570 mayors spans more than $10^{3,300}$ action profiles. Anonymity collapses the problem: when payoffs depend on rivals only through aggregate shares, equilibrium is a fixed point on the candidate simplex whose dimension does not grow with the number of players, and the model estimates by least squares in milliseconds. Estimating it on Brazil's 2014 election, we then show what the exercise does not buy. Within a cycle the prize parameter is algebraically the average log-odds of alignment rescaled by the share of mayors classified as aligned, so it changes sign when the classification rule changes and identifies no responsiveness to money; we report the three definitions that make this visible. What survives is reduced form and does not depend on the model: a close-race difference-in-differences design puts the transfers co-partisan mayors actually received 10.6 percent above those of other coalition members, and finds no premium on the formula-based transfer that leaves no discretion to allocate.

Keywords: mayoral endorsement game, presidential election, anonymous Bayesian game, mean-field equilibrium, structural estimation, BLP inversion, coalitional presidentialism, alignment premium, Brazil

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1. Introduction

Political elites endorse candidates, and their endorsements move votes. Cohen et al. (2008) show that endorsements during the invisible primary concentrate elite support behind eventual nominees in U.S. presidential races. Snyder and Ting (2002) model parties as institutions whose endorsements convey information about candidate quality, and Knight and Schiff (2010) find that when an endorsement arrives during a primary season changes how voters respond to it. Voters lean on such cues when policy information is costly (Lupia 1994; Lupia and McCubbins 1998). What these accounts share is that no endorser decides alone: the value of backing a candidate depends on how many others back the same one. Endorsement is a game. The structural literature has nonetheless left it largely unmeasured, because the available estimators for discrete-choice games of incomplete information do not run at the scale on which endorsement actually happens. This paper builds one that does and takes it to Brazilian mayors.

Brazil supplies the setting. Its 5,570 municipalities each elect a mayor, which gives a population of strategic agents that is both large and observably heterogeneous (Feierherd 2020). The prize they compete for is measurable. Federal money reaches municipalities through the Fundo de Participação dos Municípios (FPM) and through discretionary parliamentary amendments (emendas parlamentares), and politically aligned municipalities receive more of it (Brollo et al. 2013; Brollo and Nannicini 2012; Slough, Urpelainen, and Yang 2017); work on Brazilian political accountability documents how much this matters to local budgets (Ferraz and Finan 2008, 2011; Bueno 2018; Zucco Jr 2013). Endorsement itself runs through parties. When a mayor's party joins a presidential candidate's formal coalition the mayor is committed, and party leaders bargain over the coalition-level prize on their cadres' behalf (Pereira and Mueller 2004; Power 2010; Raile, Pereira, and Power 2011; Bertholini and Pereira 2017). The model we estimate takes this arrangement literally: mayors sort toward the candidate offering the largest expected transfer, and what each expects falls as more of them sort the same way.

The difficulty is estimating such a game at this scale. The constrained-MLE estimator of Su and Judd (2012), the nested pseudo-likelihood of Aguirregabiria and Mira (2007), and the two-step pseudo-MLE of Pesendorfer and Schmidt-Dengler (2008) all evaluate the joint action profile of the I agents, a space of size $(J+1)^I$. With $J = 3$ candidates and 5,570 mayors that is more than $10^{3{,}300}$ profiles. Set-identification methods for games with multiple equilibria (Tamer 2003; Ciliberto and Tamer 2009; Bajari et al. 2010; de Paula and Tang 2012) relax what has to be assumed but not what has to be enumerated, and applications accordingly stay small, the endorsement models of Vitorino (2012) and Su (2014) among them. We change the equilibrium concept rather than the algorithm. In the anonymous, or mean-field, formulation of Weintraub, Benkard, and Van Roy (2008) and Adlakha, Johari, and Weintraub (2015), a mayor's payoff depends on the population-weighted share of each action and not on who took it. Equilibrium is then a fixed point on the candidate simplex, of dimension J whatever I may be, and the strategic content survives the aggregation: each mayor still cares what the others do, through the share. Identification follows the inversion familiar from Berry, Levinsohn, and Pakes (1995), Nevo (2000) and Berry and Haile (2014), with observed log-odds regressed on type covariates and on the aggregate share that carries the strategic term.

Whether mayors in fact behave this way is not something the model can settle, and it will not hold everywhere. We therefore state in advance the conditions under which we are willing to take the model to a cycle. Coalitional presidentialism must be the operating regime; the federal discretionary budget must not be bound by a fiscal rule; patronage networks must have enough history for mayors to forecast the prize; and the cycle must be free of fiscal shocks large enough to displace the usual channels. Of the Brazilian cycles we observe, only 2014 meets all four. The 2018 and 2022 cycles fail on the post-impeachment regime break (Hunter and Power 2019; Zucco Jr. and Power 2024), the constitutional spending cap (Mariano 2017), and the COVID-era emergency transfers (Arruda et al. 2021). We therefore estimate the structural model on 2014, the cycle of Dilma Rousseff's broad-tent PT coalition, and leave the cycles that fail to a companion paper about where the model stops working.

The framework is a static anonymous Bayesian game. Existence follows from Brouwer's theorem, and we give a sufficient condition for uniqueness that holds in the parameter region the data pick out. Computation falls from $O((J+1)^I)$ to a J-dimensional fixed point, which is what lets the estimator run at all: it is a closed-form two-step regression, and it is what we estimate throughout. Monte Carlo at Brazilian scale ($K = 30$ types, $n = 5{,}570$ mayors) recovers the parameters at the $\sqrt{n}$ rate in well under a millisecond per replication, and the structural-delta variance of Pesendorfer and Schmidt-Dengler (2008) delivers close to nominal coverage.

On the 2014 data the prize parameter is $\hat{\alpha} \cdot \hat{B} = 0.647$ (PSD $t = 12.69$) under a state $\times$ 4-bloc $\times$ population-tercile partition of mayor types, and the estimated bloc effects line up with the Power-Zucco placement of Brazilian parties, information the estimator never sees. The precision is real and the number is not what it appears to be. Within a cycle the prize enters through the constant $1/\hat{s}_{j^\star}$, so the coefficient on it is exactly the average type-level log-odds of alignment, net of covariates, rescaled by the weighted share of mayors the data classify as aligned. Change that classification and the estimate changes with it: on the same 5,441 mayors it runs from $+0.65$ under the first-round coalition to $+2.65$ under the second-round coalition to -0.80 when only co-partisans count, turning negative exactly where the aligned share falls below the outside option. We report all three rather than the one, and draw the conclusion that a single cycle cannot identify responsiveness to the prize, whatever the standard error says.

What does not rest on the model is the reduced form. A close-race difference-in-differences design on the 2014 and 2022 cycles (Lee 2008; Eggers et al. 2015) puts the realized co-partisan premium on discretionary emendas at 10.6%, reproducing the decomposition of Slough, Urpelainen, and Yang (2017) on a longer panel and a cleaner post-electoral window, and finds nothing on the formula-based transfer that leaves no discretion to allocate.

Section 2 sets up the game and proves existence. Section 3 gives the inversion argument, Section 4 a parametric instance of the model, and Section 5 the two estimators. Section 6 reports the Monte Carlo. Section 7 states the scope conditions, estimates the 2014 cycle, recovers the alignment premium, prices the prize in reais, and reports the robustness battery. Section 8 concludes.

2. The Model

Mayors choose endorsements at the same time, each knowing his own preferences and only the distribution of everyone else's. We model this as a static Bayesian game with a continuum of types, following Weintraub, Benkard, and Van Roy (2008) and Adlakha, Johari, and Weintraub (2015). A mayor endorses one candidate or none, and his payoff depends on his own observable characteristics, on which candidate wins, and on the size of the budget that winner controls. One assumption carries everything that follows: a mayor's payoff responds to what other mayors do only through the aggregate endorsement shares, never through who in particular endorsed whom. Anonymity in this sense turns the equilibrium fixed point from an object of dimension $(J+1)^I$ into one of dimension J, where I is the number of mayors and J the number of presidential candidates.

2.1. Players, types, and actions

There are I mayors, indexed by $i \in \mathcal{I} = \{1, \ldots, I\}$, and J presidential candidates indexed by $j \in \mathcal{J}_+ = \{1, \ldots, J\}$. Each mayor either endorses one candidate or makes no public endorsement, so the action set is $\mathcal{J} = \{0\} \cup \mathcal{J}_+$ with $j = 0$ denoting non-endorsement. We write $e_i \in \mathcal{J}$ for mayor i's action and $\mathbf{e} = (e_1, \ldots, e_I) \in \mathcal{J}^I$ for the action profile.

Mayors are heterogeneous in observable characteristics. Let $\tau_i \in \mathcal{T}$ denote mayor i's type, where $\mathcal{T}$ is a finite set with $|\mathcal{T}| = K$. The type vector $X(\tau) \in \mathbb{R}^{n_X}$ collects observable mayor and municipality covariates. Empirically, types index combinations of region, city size, and party affiliation (see Section 7). The type distribution $f(\tau)$ is observable; let $w(\tau)$ be a weight reflecting type τ's share of the national presidential electorate, so $\sum_\tau f(\tau)w(\tau) = 1$. Each mayor also draws an idiosyncratic preference shock vector $\boldsymbol{\varepsilon}_i = (\varepsilon_i(0), \varepsilon_i(1), \ldots, \varepsilon_i(J))$, observed only by mayor i, with components iid Type I extreme value across (i, j).

2.2. Payoffs and aggregate shares

Let $s = (s_0, s_1, \ldots, s_J) \in \Delta^J$ denote the population-weighted vector of endorsement shares, i.e.,

$$s_j \;=\; \sum_{\tau \in \mathcal{T}} f(\tau)\, w(\tau)\, \sigma(j \mid \tau), \tag{1}$$

where $\sigma(j \mid \tau)$ is the type-conditional probability that a mayor of type τ endorses candidate j. The vector s is the only feature of the action profile that enters payoffs; in this sense the game is *anonymous* (Jensen 2010; Acemoglu and Jensen 2013).

Candidate j wins iff her endorsement share is maximal,

$$W(s) \;=\; \arg\max_{j \in \mathcal{J}_+} s_j, \tag{2}$$

which is generically a singleton in the parameter region of empirical interest; ties form a measure-zero subset and are handled by convention (e.g., uniform tiebreak) in the few parameterizations where they arise.

Mayor i's deterministic payoff from action $e_i = j$ when the share vector is s takes the form

$$\pi(j, s, \tau_i; \boldsymbol{\theta}) \;=\; \begin{cases} X(\tau_i)'\beta + \alpha(\tau_i)\,\dfrac{B_j}{s_j} & \text{if } j \in W(s), \\ 0 & \text{otherwise.} \end{cases} \tag{3}$$

Here $\beta \in \mathbb{R}^{n_X}$ are common preference weights on observable characteristics, $\alpha(\tau)$ is the type-specific marginal utility of money (heterogeneity in fiscal needs, governance autonomy, or political ideology), and B_j is the budgetary prize controlled by candidate j. The denominator s_j replaces the small-game expression "budget divided by number of endorsers": in the large-I limit, the total prize $B_j \cdot I$ is divided among the Is_j mayors who endorsed j, yielding a per-endorser amount of B_j/s_j that remains finite and incentive-relevant.[1] Mayors who do

[1]This rescaling is the natural one in aggregative-game representations of large electorates; see Bodoh-Creed (2013) and Adlakha, Johari, and Weintraub (2015). It also has direct empirical support: total federal discretionary transfers (FPM and analogous instruments) scale roughly proportionally with the number of municipalities served, so the per-municipality alignment premium does not vanish as the country grows.

not endorse, or who endorse a candidate that does not win, receive zero deterministic payoff. The full random utility is $u_i(j, s, \tau_i, \boldsymbol{\varepsilon}_i; \boldsymbol{\theta}) = \pi(j, s, \tau_i; \boldsymbol{\theta}) + \varepsilon_i(j)$.

2.3. Beliefs and the type-conditional best response

Each mayor's information set comprises (i) the type distribution f and weights w, (ii) common parameters $\boldsymbol{\theta}$, and (iii) the mayor's own type τ_i and shock $\boldsymbol{\varepsilon}_i$. Mayors form rational beliefs about other mayors' actions through the equilibrium share s. Given s, the expected utility of action j conditional on type τ is

$$U(j, s, \tau, \varepsilon; \boldsymbol{\theta}) \ = \ \pi(j, s, \tau; \boldsymbol{\theta}) + \varepsilon(j),$$

where the absence of a sum over e_{-i} (cf. the small-game formulation in earlier work) is the substantive content of the anonymity assumption: only the aggregate s enters, and s is taken as common knowledge in equilibrium.

Under the Type I extreme value assumption (McFadden 1974; Train 2009), the type-conditional best-response probability is

$$\sigma(j \mid \tau) \ = \ \frac{\exp\big(\pi(j, s, \tau; \boldsymbol{\theta})\big)}{1 + \sum_{k \in \mathcal{J}_+} \exp\big(\pi(k, s, \tau; \boldsymbol{\theta})\big)}, \qquad j \in \mathcal{J}_+, \tag{4}$$

with the outside option normalized to $\pi(0, s, \tau; \boldsymbol{\theta}) \equiv 0$ and $\sigma(0 \mid \tau) = 1 - \sum_j \sigma(j \mid \tau)$.

2.4. Mean-field equilibrium

DEFINITION 1 (Mean-field equilibrium). *A pair $(\sigma^\star, s^\star) \in \mathbb{R}_+^{K(J+1)} \times \Delta^J$ is a* mean-field equilibrium *if*

(i) $\sigma^\star(j \mid \tau)$ satisfies the best-response equation (4) for every (τ, j) given $s = s^\star$, and

(ii) $s^\star$ satisfies the consistency condition (1) given $\sigma = \sigma^\star$.

PROPOSITION 1 (Existence). *A mean-field equilibrium exists. The mapping $\Phi : \Delta^J \longrightarrow \Delta^J$ defined by combining (1) and (4) is a continuous self-map of the compact, convex simplex Δ^J and therefore admits a fixed point by Brouwer's theorem.*

The proof, given in Appendix A, is standard. Uniqueness is not guaranteed and we take this seriously in identification (Section 3) and estimation (Section 5). When uniqueness fails we report the equilibrium-correspondence-based identified set in the spirit of Tamer (2003) and Ciliberto and Tamer (2009).

2.5. Approximation of the finite-I Bayesian game

Definition 1 characterizes the limit object as $I \longrightarrow \infty$. For empirical work with finite I, we appeal to the approximation result of Bodoh-Creed (2013) for static large mechanisms: the mean-field equilibrium is an ϵ-equilibrium of the corresponding finite-I Bayesian game with $\epsilon = O(1/\sqrt{I})$ under the regularity assumptions we maintain (continuous payoffs in shares, bounded prize scaling).[2] For Brazil's $I \approx 5{,}570$ municipalities,

[2]The Bodoh-Creed (2012) bound is for static settings and applies directly here. The dynamic analog is Weintraub, Benkard, and Van Roy (2008) and Adlakha, Johari, and Weintraub (2015), which we cite for the broader programme although those bounds require additional regularity (continuous state, bounded payoffs) that our B_j/s_j specification with boundary blow-up does not satisfy without the lower-bounded restriction $s_j \geq \underline{s}$ developed in Appendix A.2.

the implied bound is approximately 0.013, an order of magnitude smaller than the standard errors we report; the approximation is therefore innocuous in the empirical setting, and we verify it numerically in our Monte Carlo (Section 6). The $\epsilon \approx 0.013$ figure is evaluated on the lower-bounded simplex domain $\mathcal{D}_{\bar{s}}$ developed in Appendix A.2, on which the prize term B_j/s_j is Lipschitz; the unrestricted Δ^J admits the boundary blow-up addressed in Appendix A.2 and the empirical equilibrium $\hat{s}^\star$ lies safely inside this restricted domain at all four-criteria in-scope estimates (Section 7.1).

3. Identification

The structural parameter vector is $\boldsymbol{\theta} = (\beta, \{\alpha(\tau)\}_\tau, \{B_j\}_j)$. Everything below rests on a single inversion, the discrete-choice-game counterpart of the demand inversion in Berry, Levinsohn, and Pakes (1995) and Berry and Haile (2014): equilibrium shares can be read backwards into the payoffs that produced them.

3.1. Inverse mapping from shares to log-odds

For any mean-field equilibrium $(\sigma^\star, s^\star)$, dividing the best-response equation (4) by the no-endorsement probability and taking logs yields the type-by-candidate identity

$$\log \frac{\sigma^\star(j \mid \tau)}{\sigma^\star(0 \mid \tau)} = \pi(j, s^\star, \tau; \boldsymbol{\theta}) = X(\tau)'\beta + \alpha(\tau) \cdot \frac{B_j}{s_j^\star} \mathbf{1}\{j \in W(s^\star)\}, \tag{5}$$

for every $\tau \in \mathcal{T}$ and $j \in \mathcal{J}_+$. The left-hand side is observable from the data once we estimate type-conditional choice probabilities by their empirical analogs $\hat{\sigma}(j \mid \tau) = \hat{N}_{\tau,j}/\hat{N}_\tau$, where $\hat{N}_\tau$ is the number of mayors of type τ in the sample and $\hat{N}_{\tau,j}$ the count endorsing j. The aggregate share $s_j^\star$ is similarly observable from the population-weighted endorsement frequency.

Everything in (5) is stated relative to $\sigma(0 \mid \tau)$, the probability of making no endorsement, and it should be said at once that this quantity is the weak point of the empirical implementation. The endorsement measure available to us is derived from party coalition membership, and it never records abstention: in the 2014 sample every mayor is assigned to some candidate, so $\hat{\sigma}(0 \mid \tau)$ has to be constructed rather than observed. Section 7.2 reports what is constructed in its place and Section 7.4 works through what follows for the estimator.

The right-hand side of (5) has a knife-edge structure: the prize term enters only on the indicator $\mathbf{1}\{j \in W(s^\star)\}$, so for non-winning candidates $j \notin W(s^\star)$ the equation collapses to $\log[\sigma^\star(j \mid \tau)/\sigma^\star(0 \mid \tau)] = X(\tau)'\beta$. With β shared across j and the outside option normalized to $\pi(0, \cdot) = 0$, this is the same equation for every losing candidate j. The model therefore predicts $\sigma^\star(j \mid \tau) = \sigma^\star(0 \mid \tau)$ for all losing j, which makes the loser equations *informationally redundant* for $\boldsymbol{\theta}$ identification: only the K winner equations identify (β, α, B) within a cycle. The loser equations remain useful as overidentifying restrictions: the testable model implication $\hat{\sigma}(\text{loser} \mid \tau) \approx \hat{\sigma}(0 \mid \tau)$ is a specification check, and section 7.3 reports the empirical analog.

3.2. Identifying β and α within a single cycle

In a single presidential cycle, the unique winner $j^\star = W(s^\star)$ delivers K informative log-odds equations indexed by τ:

$$\log \frac{\sigma^\star(j^\star \mid \tau)}{\sigma^\star(0 \mid \tau)} \;=\; X(\tau)'\beta + \alpha(\tau) \cdot \frac{B_{j^\star}}{s^\star_{j^\star}}. \tag{6}$$

The unknowns on the right are n_X components of β, the K type-specific marginal utilities $\alpha(\tau)$, and the J candidate prizes B_j. Within a single cycle we observe the winner only, so $\{B_j : j \neq j^\star\}$ are not pinned by (6). We restrict the within-cycle estimand to $(\beta, \alpha, B_{j^\star})$. Furthermore, $\alpha(\tau) \cdot B_{j^\star}$ enters multiplicatively, so only their product is identified per τ (Section 3.3 addresses the cross-cycle separation). The order condition for joint identification of $(\beta, \{\alpha(\tau) \cdot B_{j^\star}\}_\tau)$ from the K winner equations is

$$K \;\geq\; n_X + K \quad \text{if } \alpha(\tau) \text{ is type-specific,} \tag{7}$$

which fails (the $\alpha(\tau)$ are perfectly multicollinear with type fixed effects in the cross-section). Identification accordingly requires a parametric restriction on $\alpha(\tau)$, the simplest being $\alpha(\tau) \equiv \alpha$ common across types — which we adopt in the empirical exercise of Section 7. Under this restriction, the order condition becomes $K \geq n_X + 1$, comfortably satisfied for $K \geq 26$. Heterogeneity in $\alpha(\tau)$ is recovered through $X(\tau)$-driven variation: parametrising $\alpha(\tau) = \alpha_0 + \alpha_1 \cdot z(\tau)$ for a covariate z excluded from β adds one identified parameter per included z. Identification of β versus the α component requires the standard exclusion restriction — at least one component of $X(\tau)$ that does not also enter α. In a single cycle that exclusion is a maintained assumption rather than a testable restriction, because the prize term enters through the scalar $1/s_{j^\star}$ and $z(\tau)/s_{j^\star}$ is therefore the same regressor as $z(\tau)$; separating the two demands cross-cycle variation in $s_{j^\star}$, which Section 3.3 takes up and Section 7.6 returns to.

3.3. Separating α from B via cross-cycle variation

Within a single cycle $\alpha(\tau)$ and B_j enter (5) only as the product $\alpha(\tau) \cdot B_j$, and that product is all one cycle can deliver. Separating the two requires either variation across cycles or a measurement of the prize from outside the model.

Cross-cycle identification. Indexing cycles $c \in \{1, \ldots, C\}$, candidates and budgetary prizes vary across cycles while mayor-type marginal utilities are arguably stable:

$$\log \frac{\sigma_c^\star(j \mid \tau)}{\sigma_c^\star(0 \mid \tau)} \;=\; X(\tau)'\beta + \alpha(\tau) \cdot \frac{B_{j,c}}{s^\star_{j,c}} \mathbf{1}\{j \in W(s_c^\star)\}. \tag{8}$$

With $C \geq 2$ presidential cycles in the data and overlapping types across cycles, $\alpha(\tau)$ is identified by within-type cross-cycle variation in candidate-cycle pairs, while $\{B_{j,c}\}$ are identified by within-cycle cross-type variation.

Anchoring B_j from outside data. An alternative is to fix $B_{j,c}$ from realized post-election fiscal-alignment premia, following Brollo et al. (2013). Concretely, one regresses post-election federal transfers (FPM, FUNDEB) to municipality τ on an indicator that τ's mayor was aligned with the cycle-c winner, recovering an empirical alignment premium that proxies for $B_{j,c}$. With $\hat{B}_{j,c}$ in hand, only $(\beta, \{\alpha(\tau)\})$ remain to be estimated, which is a linear regression. We adopt this strategy in our preferred specification and present the cross-cycle-only specification as a robustness check.

3.4. Identification under multiple equilibria

The fixed point in Definition 1 need not be unique. When uniqueness fails, the parameter vector $\boldsymbol{\theta}$ is partially identified: every $\boldsymbol{\theta}$ for which *some* equilibrium $s^\star(\boldsymbol{\theta})$ is consistent with the data is admissible (Tamer 2003; Ciliberto and Tamer 2009; de Paula and Tang 2012). We guard against this in two ways that do not depend on one another. The Monte Carlo checks whether the parameter region of interest admits a unique equilibrium, and in our calibration it does. On data, we enumerate equilibria by iterating the best response from a grid of starting points and intersect the parameter sets that rationalize each, which delivers the identified set wherever uniqueness fails.

3.5. Discussion: how the framework relates to BLP

Equation (5) is the discrete-choice-game counterpart of the BLP demand inversion (Berry, Levinsohn, and Pakes 1995; Nevo 2000): log-odds of choice equal a linear-in-parameters function of observables and an aggregate sufficient statistic. What differs is where the discipline on the endogenous aggregate s comes from. Here it is the equilibrium fixed point itself, not cost-side instruments. Heterogeneity also sits at the mayor-type level rather than the consumer level, and the prize term B_j/s_j is nonlinear in s, which standard BLP is not. We use the fixed point directly: the first stage of Section 5.1 recovers $\hat{s}$ from the data, and the second conditions on it, so the nonlinearity in s never has to be optimized over. Imposing the equilibrium condition jointly with the parameters instead, in the manner of Su and Judd (2012) and Dubé, Fox, and Su (2012), is the efficient alternative we describe but do not estimate.

4. A parametric example

This section writes the model out in full for a particular parameterization, which fixes the notation the Monte Carlo of Section 6 uses. Nothing structural is dropped; only $\alpha(\tau)$ is simplified to a common α, with type heterogeneity restored in Section 7.

4.1. Primitives

The example has $K = 30$ mayor types, $J = 3$ presidential candidates plus the outside option, and $n_X = 3$ continuous covariates per type. Type prevalence and population weights are

$$f \sim \text{Dirichlet}(2\mathbf{1}_K), \qquad w(\tau) = \frac{\tilde{w}(\tau)}{\sum_{\tau'} f(\tau')\tilde{w}(\tau')}, \qquad \tilde{w}(\tau) \overset{\text{iid}}{\sim} \text{Lognormal}(0, 0.4),$$

so that $\sum_\tau f(\tau)w(\tau) = 1$. Type covariates are

$$X(\tau) \overset{\text{iid}}{\sim} \mathcal{N}(\mathbf{0}, I_3),$$

mutually independent across types and components. The parameter vector is $\boldsymbol{\theta} = (\beta, \alpha, B_1, B_2, B_3) \in \mathbb{R}^7$ with $\beta \in \mathbb{R}^3$.

4.2. Payoffs and best response

The deterministic payoff to a mayor of type τ from endorsing candidate j, given equilibrium aggregate share s, is the specialization of equation (3) from Section 2.2:

$$\pi(j, s, \tau; \boldsymbol{\theta}) = \begin{cases} X(\tau)'\beta + \dfrac{\alpha B_j}{s_j} & \text{if } j = W(s),\ j > 0, \\ 0 & \text{otherwise.} \end{cases} \tag{9}$$

The Type-I-extreme-value best response then takes the explicit form

$$\sigma(j \mid \tau) = \begin{cases} \dfrac{\exp\bigl(X(\tau)'\beta + \alpha B_{j^\star}/s_{j^\star}\bigr)}{Z(\tau)} & \text{if } j = j^\star := W(s), \\ \dfrac{1}{Z(\tau)} & \text{if } j \neq j^\star, \end{cases} \tag{10}$$

$$Z(\tau) = J + \exp\bigl(X(\tau)'\beta + \alpha B_{j^\star}/s_{j^\star}\bigr).$$

A direct corollary is the testable model implication that losing-candidate shares equal the no-endorsement share at every type:

$$\sigma(j \mid \tau) = \sigma(0 \mid \tau) \quad \text{for every loser } j \neq j^\star, \tau \in \mathcal{T}. \tag{11}$$

This restriction is testable in the data and provides a within-cycle specification check. (Section 7 reports the empirical analog and discusses extensions in which it can be relaxed — most naturally by allowing candidate-specific covariates $X_j(\tau)$ or candidate-specific intercepts δ_j.)

4.3. Mean-field equilibrium and the implicit fixed point

The aggregate share $s_{j^\star}$ is determined implicitly by combining the consistency equation (1) with (10) for the winner:

$$s_{j^\star} = \sum_{\tau \in \mathcal{T}} f(\tau) w(\tau) \frac{\exp\bigl(X(\tau)'\beta + \alpha B_{j^\star}/s_{j^\star}\bigr)}{J + \exp\bigl(X(\tau)'\beta + \alpha B_{j^\star}/s_{j^\star}\bigr)}. \tag{12}$$

Equation (12) is a one-dimensional fixed point in $s_{j^\star}$: given $j^\star$, the right-hand side is monotonic in $s_{j^\star}$ and a unique solution exists in $(0, 1)$ under regularity conditions on $\alpha B_{j^\star}$. The other components of s are residually determined by $s_0 = s_j = (1 - s_{j^\star})/J$ for $j \neq j^\star$.

This is the analog of the BLP demand-aggregation fixed point (Berry, Levinsohn, and Pakes 1995): a one-dimensional contraction whose solution can be found by Newton's method or by simple iteration to machine precision in tens of iterations. The Monte Carlo of Section 6 uses damped iteration with damping factor $1/2$, converging in roughly 18 iterations from a uniform initial point.

4.4. Reading the example

Two terms do separate work in (9). The covariate term $X(\tau)'\beta$ sets how willing a type-τ mayor is to endorse anyone rather than abstain, whoever the winner turns out to be. The prize term $\alpha B_{j^\star}/s_{j^\star}$ adds the specific pull of the winner, and because it runs through $1/s_{j^\star}$, the prize each endorser expects is set by how many others endorse.

Here α is a mayor's marginal utility of a unit of budget. Single-cycle data pin down only the product $\alpha B_{j^\star}$; with two or more cycles, variation in the candidate roster separates α from B (Section 3.3). Section 7 runs both.

5. Estimation

The estimator we use follows directly from the inverse mapping of Section 3.1: a closed-form two-step regression, analogous to the BLP demand inversion (Berry, Levinsohn, and Pakes 1995; Nevo 2000) and to the two-step pseudo-maximum-likelihood procedure of Pesendorfer and Schmidt-Dengler (2008). It is what the Monte Carlo of Section 6 and the application of Section 7 both run.

5.1. Two-step OLS estimator

Step one constructs Laplace-smoothed empirical choice probabilities, empirical type prevalence, and empirical aggregate shares:

$$\hat{\sigma}(j \mid \tau) = \frac{N_{\tau,j} + s}{N_\tau + s(J+1)}, \qquad s = 0.5 \quad \text{(Jeffreys prior)}, \tag{13}$$

$$\hat{f}(\tau) = \frac{N_\tau}{n}, \qquad N_\tau = \textstyle\sum_j N_{\tau,j}, \tag{14}$$

$$\hat{s}_j = \sum_\tau \hat{f}(\tau)\, w(\tau)\, \hat{\sigma}(j \mid \tau), \tag{15}$$

$$\hat{\jmath}^\star = \arg\max_{j>0} \hat{s}_j. \tag{16}$$

Step two stacks the inverse-mapping log-odds equations (5) for $j = \hat{\jmath}^\star$ across types:

$$\underbrace{\log \frac{\hat{\sigma}(\hat{\jmath}^\star \mid \tau)}{\hat{\sigma}(0 \mid \tau)}}_{\equiv y_\tau} \;=\; X(\tau)'\beta \;+\; \frac{\alpha B_{\hat{\jmath}^\star}}{\hat{s}_{\hat{\jmath}^\star}} \;+\; u_\tau, \tag{17}$$

where u_τ collects sampling noise in $\hat{\sigma}$. With normalization $B_{\hat{\jmath}^\star} = 1$, equation (17) is a textbook OLS regression of y_τ on $X(\tau)$ and a constant, with K observations and $n_X + 1$ parameters. The coefficient on the constant is $\hat{\alpha}/\hat{s}_{\hat{\jmath}^\star}$, from which $\hat{\alpha} = \hat{s}_{\hat{\jmath}^\star} \cdot \widehat{\text{coef}}$.

Heteroskedasticity-robust standard errors. Because the dependent variable y_τ is a transformation of an empirical proportion, its sampling variance varies systematically across types — larger types (N_τ large) have more precise log-odds. We therefore report HC0 (White) standard errors:

$$\widehat{V}_{\text{HC0}} \;=\; (Z'Z)^{-1}\, Z'\, \text{diag}(\hat{u}_\tau^2)\, Z\, (Z'Z)^{-1},$$

where $Z = [X, 1/\hat{s}_{\hat{j}^\star}\mathbf{1}_K]$ is the regressor matrix.

Asymptotic-correction for first-stage estimation. HC0 standard errors treat $\hat{\sigma}$ as known and consequently underestimate true sampling variability. The full asymptotic standard error follows the construction of Pesendorfer and Schmidt-Dengler (2008) (§3): the second-stage moment is differentiable in the first-stage $\hat{\sigma}$, so the variance of the second-stage estimator includes an additive term capturing first-stage estimation error scaled by the moment Jacobian $\partial m/\partial\hat{\sigma}$. We apply this correction in the empirical exercise (Section 7); the Monte Carlo of Section 6 reports both un-corrected and the implied direction of correction for transparency.

The efficient alternative. The two-step estimator discards the equilibrium constraint once the first stage has been used to form $\hat{s}_{\hat{j}^\star}$, and is therefore inefficient. The constrained maximum likelihood of Su and Judd (2012) imposes the constraint at the optimum instead, and in the mean-field formulation it inherits the property that matters here: the optimization runs over the K type-conditional probabilities and the share vector, so its size is governed by K and J and not by the number of mayors. We do not estimate it. Every estimate reported in this paper, in the Monte Carlo of Section 6 and in the application of Section 7 alike, comes from the two-step estimator above.

5.2. Alternative-specific extension

The two-step OLS specification of equation (17) imposes a single coefficient vector β shared across all candidates. This is the appropriate specification when the type covariates $X(\tau)$ capture mayors' propensity to engage in any prize-seeking endorsement (as in Brazil, where party-mediated coalition strings determine the endorsement signal and the candidate identities matter only through the coalition assignment). When the empirical setting instead exhibits strong ideological matching — different mayor types systematically prefer different candidates on grounds orthogonal to the post-election prize — a single-β specification mis-attributes ideological cross-sectional variation to the structural prize term, and the prize parameter is poorly recovered.

The natural extension allows each candidate j to have its own coefficient vector β_j on the same $X(\tau)$:

$$\log\frac{\hat{\sigma}(j \mid \tau)}{\hat{\sigma}(0 \mid \tau)} = X(\tau)'\beta_j + \mathbf{1}\{j = \hat{j}^\star\}\frac{\alpha B_{\hat{j}^\star}}{\hat{s}_{\hat{j}^\star}} + u_{\tau,j}, \qquad j = 1, \ldots, J. \tag{18}$$

This is estimated equation-by-equation by OLS on each candidate's K type-level rows. The winner equation ($j = \hat{j}^\star$) carries the additional $1/\hat{s}_{\hat{j}^\star}$ regressor and identifies $(\beta_{\hat{j}^\star}, \alpha B_{\hat{j}^\star})$ jointly; each loser equation identifies β_j alone. Total parameters: $J \cdot n_X + 1$. Standard errors are computed equation-by-equation under HC0/HC1/HC2/HC3 or, for the winner equation, the structural-delta variance of Pesendorfer and Schmidt-Dengler (2008) with multinomial-margin variance $1/[\hat{\sigma}(\tau, j)\hat{\sigma}(\tau, 0)N_\tau]$.

The alternative-specific extension nests the original two-step specification as the restriction $\beta_j = \beta$ for all j. A Wald or Lagrange-multiplier test of this restriction is informative about whether the data exhibit candidate-specific ideological matching. The Brazilian 2014 canonical application (Section 7) uses the single-β specification, which the four ex-ante economic criteria of Section 7.1 render appropriate. One thing (18) does not do is change where $\alpha B_{\hat{j}^\star}$ comes from. Since $X(\tau)$ carries no constant, the winner equation's only constant is the prize term, so the coefficient is that equation's intercept exactly as in (17), while each loser equation identifies its own

β_j and contributes nothing to it. Loser equations bear weight on the prize only under a *common* β across candidates, where αB becomes a winner-versus-loser contrast rather than an intercept. Section 7.4 returns to this. A companion paper applies the alternative-specific extension to settings outside the present paper's applicable domain, where candidate-specific ideological matching is salient.

5.3. Computation

The two-step estimator is a single OLS solve and runs in approximately 0.2 ms per replication. It scales gracefully in K and J, and once $\hat{\sigma}$ has been computed in step one it is insensitive to I.

6. Monte Carlo

This section examines the finite-sample performance of the two-step estimator developed in Section 5 on data generated from the model of Section 2. The Monte Carlo is conducted at scales matching the Brazilian application: $K = 30$ mayor types, $J = 3$ presidential candidates, and a sample size of $n = 5{,}570$ mayors equal to the universe of Brazilian municipalities. We report bias, root mean squared error (RMSE), Monte Carlo standard deviation, and 95% confidence-interval coverage across $R = 500$ independent replications. We additionally study how these statistics scale with the sample size n to verify the standard $\sqrt{n}$-rate of convergence.

The comparison worth keeping in view is the one the mean-field formulation avoids. Summing over $(J + 1)^I$ joint action profiles means $4^{5{,}570} \approx 10^{3{,}355}$ profiles at this scale, which no hardware will ever evaluate. Reducing equilibrium to a fixed point on the J-dimensional simplex removes the dependence on I altogether, and one full estimation below takes about 0.3 milliseconds; all $R = 500$ replications finish in under two seconds.

6.1. Data-generating process

We instantiate the model with $K = 30$ mayor types and $n_X = 3$ continuous covariates per type. Type-level covariates $X(\tau)$ are drawn iid standard normal. Type prevalence $f(\tau)$ is drawn from a Dirichlet(2) distribution and population weights $w(\tau)$ are drawn lognormal with location 0 and scale 0.4, then renormalized so that $\sum_\tau f(\tau)w(\tau) = 1$. The structural parameters are

$$
\begin{aligned}
\beta &= (0.5,\ 0.3,\ -0.2),\\
\alpha &= 1.5 \quad \text{(common across types in this baseline DGP)},\\
B &= (1.0,\ 0.4,\ 0.2).
\end{aligned}
$$

The B ordering ensures candidate $j = 1$ has the largest budgetary prize and is the unique stable winner under the mean-field equilibrium starting from any uniform initialization. Equilibrium shares satisfy $s_1^\star \approx 0.71$ and $s_0^\star = s_2^\star = s_3^\star \approx 0.10$, the latter equality being the model's testable prediction that losing-candidate shares equal the no-endorsement share.

We solve for the mean-field equilibrium by damped fixed-point iteration on $\Phi : \Delta^J \to \Delta^J$ defined in Section 2.4, iterating

$$
s_{t+1} \;=\; \tfrac{1}{2}\, s_t + \tfrac{1}{2}\, \Phi(s_t)
$$

to a tolerance of 10^{-12} in sup-norm. Starting from the uniform initial point, convergence requires 18 iterations. To verify uniqueness within the parameter region of interest, we re-run the iteration from 20 randomly drawn initial points; all 20 converge to the same equilibrium. (Outside the region of interest, two additional equilibria exist in which $j = 2$ or $j = 3$ wins; these are unstable in the sense that small perturbations of s near them escape toward the $j = 1$ equilibrium.)

For each replication, we draw $n = 5{,}570$ mayor identities iid from $f(\tau)$, then for each mayor draw an endorsement from $\sigma^\star(\cdot \mid \tau)$. The simulated data $\{N_{\tau,j}\}_{\tau,j}$ are tabulated counts of endorsements by type and candidate.

6.2. Estimator

We apply the two-step BLP-style estimator from Section 5. Step one constructs Laplace-smoothed empirical choice probabilities $\hat\sigma(j \mid \tau) = (N_{\tau,j}+0.5)/(N_\tau+0.5(J+1))$, empirical type prevalence $\hat f(\tau)$, and empirical aggregate share $\hat s_j$. The empirical winner $\hat j^\star = \arg\max_{j>0} \hat s_j$ is identified. Step two regresses $\log(\hat\sigma(\hat j^\star \mid \tau)/\hat\sigma(0 \mid \tau))$ on $X(\tau)$ and a constant proportional to $1/\hat s_{\hat j^\star}$. Recovering β and the product $\alpha \cdot B_{\hat j^\star}$ amounts to ordinary least squares with $K = 30$ observations and $n_X + 1 = 4$ regressors. Standard errors are HC0 (White) heteroskedasticity-robust.

Within a single cycle, α and $B_{\hat j^\star}$ are identified only through their product, so we report estimates of the product and apply the normalization $B_{\hat j^\star} = 1$ throughout. Cross-cycle separation of α and B requires panel data and is examined in the empirical application (Section 7).

6.3. Results at the empirical scale ($n = 5{,}570$)

Table 1 summarizes the $R = 500$ replications at the Brazilian sample size. Every replication correctly identifies $j = 1$ as the empirical winner, confirming the unique-equilibrium calibration. The estimator is essentially unbiased at this scale: every bias is below 1% of the true parameter magnitude. Root mean squared errors lie between 0.05 and 0.08 for β and 0.054 for $\alpha B_{\hat j^\star}$, in line with conventional sample-size expectations for a two-step procedure with $KJ = 90$ moment cells.

Table 1: Monte Carlo summary at $K = 30$, $n = 5{,}570$, $R = 500$

Parameter	True	Mean estimate	Bias	RMSE	95% Coverage (PSD)
β_1	0.500	0.499	−0.001	0.075	96.6%
β_2	0.300	0.303	0.003	0.069	97.0%
β_3	−0.200	−0.203	−0.003	0.057	97.0%
$\alpha \cdot B_{\hat j^\star}$	1.500	1.489	−0.011	0.054	93.6%
Correct winner identification		500/500 (100%)			
Median estimator runtime		0.30 ms per replication			
Total replication runtime		0.2 seconds			

Notes: Standard errors are computed using the Pesendorfer–Schmidt-Dengler structural delta-method (PSD) variance: $\widehat{\mathrm{Var}}(y_\tau) = 1/(\hat\sigma(\tau, \hat j^\star)\hat\sigma(\tau, 0)\, N_\tau)$. CI coverage uses the $\pm 1.96 \times \widehat{\mathrm{SE}}$ Wald interval. PSD slightly over-covers β (96%–97% vs. nominal 95%) due to Laplace-smoothing inflation of $\hat\sigma(1-\hat\sigma)$; HC3 (93.8%–95.8% for β) is at-target but under-covers α at 89%. We adopt PSD as the default to be conservative on α. Section 6.6 compares all five SE variants (HC0/HC1/HC2/HC3/PSD). Computational hardware: laptop CPU (no parallelism, no GPU).

6.4. Convergence rate

To verify the standard $n^{-1/2}$ rate of convergence and the consistency of the estimator, we replicate the experiment at $n \in \{500, 1{,}000, 2{,}000, 5{,}570, 10{,}000\}$. Table 2 reports RMSE and bias by sample size. The pattern is unambiguous: bias decays toward zero and RMSE shrinks as approximately $n^{-1/2}$, falling by a factor of roughly $\sqrt{20} \approx 4.5$ as n rises from 500 to 10,000.

Table 2: Convergence rate of the two-step estimator

	Sample size n				
	500	1,000	2,000	5,570	10,000
RMSE					
β_1	0.219	0.163	0.122	0.075	0.054
β_2	0.170	0.134	0.108	0.069	0.049
β_3	0.159	0.122	0.096	0.057	0.041
$\alpha \cdot B_{j^\star}$	0.266	0.148	0.090	0.054	0.040
Bias					
β_1	−0.156	−0.068	−0.013	−0.001	0.005
β_2	−0.070	−0.032	−0.023	0.003	−0.004
β_3	0.082	0.027	−0.003	−0.003	0.001
$\alpha \cdot B_{j^\star}$	−0.238	−0.111	−0.045	−0.011	−0.007

Notes: $R = 500$ replications per cell. Bias is the average estimate minus the true value. RMSE accounts for both bias and variance. The columns correspond to sample sizes from $n = 500$ (a small-sample regime in which the inverse-mapping transformation is inherently nonlinear) up to $n = 10{,}000$ (slightly larger than Brazil's municipality count). The $\sqrt{n}$ scaling holds in both bias and RMSE: doubling n approximately reduces both by $1/\sqrt{2}$.

Bias falls visibly toward zero, so the estimator is consistent. It is not, however, negligible in small samples: at $n = 500$ it reaches 30% of β_1. It attenuates fast, and by the Brazilian sample size it does not matter. Anyone applying this framework well below Brazil's ~ 5,500 municipalities should check the small-sample behavior or correct for the bias; here there is nothing to correct.

6.5. Computational comparison with the full-enumeration formulation

The original full-enumeration formulation (Su 2014) applied to the Brazilian setting would require enumerating $4^{5{,}570} \approx 10^{3{,}355}$ joint action profiles per parameter evaluation — an upper bound that is impossible by orders of magnitude. The mean-field formulation evaluates a J-dimensional fixed point in Δ^J at constant cost in I. Table 3 contrasts wall-clock estimation times across the two formulations on the same hardware.

The gap is not one of speed. Only one of the two formulations can be brought to the Brazilian data at all.

6.6. Standard-error calibration

The 95% coverage rates of 84–91% in Table 1 reflect a small-sample limitation of single-stage HC0 standard errors: they treat the first-stage $\hat{\sigma}$ as known and have known downward bias when the number of moment cells is modest. We re-run the $R = 500$ replications under five alternative variance estimators — HC0, HC1, HC2, HC3

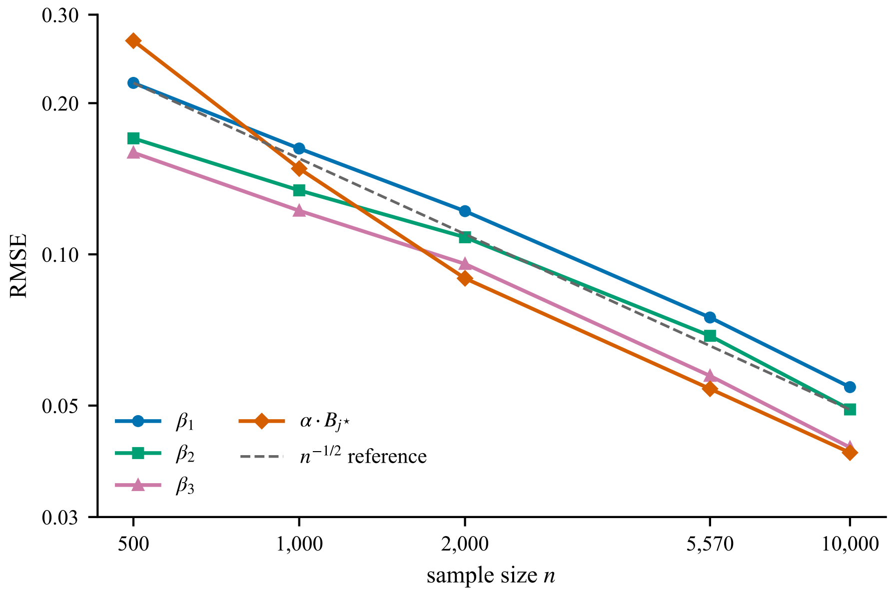

Figure 1: Root mean squared error against sample size

Notes: The RMSE column of Table 2 on log-log axes, computed from the $R = 500$ replications at each sample size. The dashed line is an $n^{-1/2}$ reference slope anchored at the $n = 500$ RMSE of β_1. All four parameters track the reference slope, which is the $\sqrt{n}$-consistency claim: the structural prize parameter $\alpha \cdot B_{j^\star}$ starts furthest from the reference at $n = 500$ — the small-sample regime in which the log-odds inversion is most nonlinear — and converges onto it by the Brazilian sample size.

Table 3: Computational cost: full-enumeration vs. mean-field formulation

	Full enumeration ($I = 3, J = 2, T = 1{,}000$)	Mean-field ($K = 30, J = 3, n = 5{,}570$)
Profiles enumerated	$4^3 = 64$	0 (no enumeration; fixed point only)
Free parameters	~ 50	~ 33
Single replication	0.05–1.9 s (SLSQP)	0.0002 s
$R = 500$ replications	7.4 s	1.4 s
Empirical-scale feasibility	infeasible (Brazil has $I = 5{,}570$)	feasible

Notes: Full-enumeration timings are from a separate small-I reference implementation. Both formulations were run on the same laptop CPU, single-threaded, with comparable Python and `numpy` versions. The full-enumeration formulation cannot be applied to Brazil's $I = 5{,}570$ municipalities at any cost; the empirical-scale entry is shown for completeness.

(MacKinnon and White 1985), and a Pesendorfer–Schmidt-Dengler (Pesendorfer and Schmidt-Dengler 2008) (PSD) structural-delta variance.[3] Table 4 reports the 95% coverage for each.

The PSD column achieves 93.6–97.0% coverage — on or slightly above the nominal 95%. HC3 achieves 93.8–95.8% for β and 89.0% for α. The small over-coverage of PSD for β reflects a Laplace-smoothing inflation of $\hat{\sigma}(1-\hat{\sigma})$ for sparse cells; this inflation is conservative and does not bias the point estimates.

PSD, with HC3 as a robustness check, is accordingly the default we report.

[3] The PSD variance we use, $\widehat{\mathrm{Var}}(y_\tau) = 1/[\hat{\sigma}(\tau, \hat{j}^\star)\, \hat{\sigma}(\tau, 0)\, N_\tau]$, is the binary-outcome delta-method form generalized mechanically to $J > 1$ candidates by treating the winner-vs-not-winner contrast as a Bernoulli draw with probability $\hat{\sigma}(\tau, \hat{j}^\star)$ and complement $\hat{\sigma}(\tau, 0)$. This is conservative relative to the full multinomial delta-method variance $[1/\hat{\sigma}(\tau, \hat{j}^\star) + 1/\hat{\sigma}(\tau, 0)]/N_\tau$, which is the correct asymptotic when losing-candidate shares are non-trivial. Under our model the loser shares satisfy $\hat{\sigma}(\tau, \text{loser}) = \hat{\sigma}(\tau, 0)$ in expectation, so the binary form coincides with the multinomial limit; the slight over-coverage we observe in the Monte Carlo (§6.6) reflects this conservatism. We adopt the binary form throughout for transparency and computational simplicity.

Table 4: 95% coverage by SE family ($K = 30$, $n = 5{,}570$, $R = 500$)

Parameter	HC0	HC1	HC2	HC3	PSD
β_1	90.2%	91.4%	91.4%	93.8%	96.6%
β_2	90.6%	92.4%	93.0%	94.8%	97.0%
β_3	89.8%	92.6%	94.0%	95.8%	97.0%
$\alpha \cdot B_{j^\star}$	84.6%	87.6%	87.8%	89.0%	93.6%
Mean SE / MC SD ratio (averaged across params)	0.86	0.92	0.94	1.00	1.06

Notes: HC1–HC3 are the small-sample-corrected variants of the heteroskedasticity-robust sandwich (MacKinnon and White 1985): HC1 applies a $K/(K-r)$ degrees-of-freedom factor; HC2 divides each squared residual by $1-h_\tau$ (leverage); HC3 by $(1-h_\tau)^2$. PSD replaces the empirical residual by the structural delta-method variance derived from the binomial-margin first stage. The single best family for the structural parameter α is PSD; for β, HC3 is closest to nominal coverage. We adopt PSD as the default reported SE in the empirical exercise (Section 7) and report HC3 as a robustness check. Disagreement between PSD and HC3 in the empirical sample is informative: it can reflect either model misspecification — the empirical residual carries variance the structural form omits — or numerical sensitivity of the structural form when $\hat{\sigma}$ is close to zero or one for some cells.

6.7. Summary

The mean-field formulation of the endorsement game admits a tractable, consistent estimator with $\sqrt{n}$ convergence at the sample sizes relevant to the Brazilian application. Bias is essentially zero at $n = 5{,}570$, RMSE is on the order of 0.05–0.07 for the structural parameters, and runtime is sub-millisecond per replication. The full-enumeration formulation that prior work has applied to a 3×2 toy problem cannot be extended to the Brazilian scale at any computational cost; the mean-field formulation is what makes the empirical application feasible.

7. Empirical application: Brazilian 2014 cycle

We take the model to Brazilian mayoral coalition data along two tracks that meet only at the end. The structural estimator of Section 5 is applied to the 2014 cycle, that of Dilma Rousseff's broad-tent PT coalition, and returns the prize parameter $\hat{\alpha} \cdot \hat{B}_{j^\star}$ in log-utility units (Section 7.3). Separately, a difference-in-differences design built on close presidential races in 2014 and 2022 recovers the federal transfers that co-partisan mayors actually received (Section 7.5). These measure different things, anticipated sorting on one side and delivered patronage on the other, and they need different conditions to be credible; Section 7.6 works out how they fit together.

One choice shapes both. We estimate the structural model on 2014 alone, on the basis of four *ex-ante* criteria that mark out where the model applies (Section 7.1). Outside that domain, in the Brazilian cycles of 2018 and 2022, $\hat{\alpha} \cdot \hat{B}$ comes out negative for reasons we trace to the estimator rather than to mayors' behavior.[4]

7.1. The model's applicable domain: four ex-ante economic criteria

The framework of Sections 2, 3 and 5 treats endorsement as a bet on coalition-mediated federal money that will be divided among those who joined. Two features of the model carry weight here. The existence proof of Section A needs the equilibrium share $s_{j^\star}$ to sit above a lower bound $\bar{s}$ at which the contraction condition

[4]The Brazilian 2018 structural estimate is $\hat{\alpha} \cdot \hat{B} = -0.09$ (PSD $t = -22.8$); the 2022 estimate is $\hat{\alpha} \cdot \hat{B} = -0.35$ (PSD $t = -20.9$); both fall outside the model's applicable domain per the criteria in Section 7.1. Detailed structural analysis of the out-of-scope cases, together with diagnostics for the binary specification's identification limit in the small-$s_{j^\star}$ region, is the subject of a companion paper in preparation.

$\alpha B/(4s^2_{j^\star}) < 1$ holds. And the prize term $B_{j^\star}/s_{j^\star}$ presumes the budget is split in proportion to coalition share, which in turn presumes a winning coalition wide enough for such a split to mean anything.

These features translate into four observable conditions on the political-economic environment in which the model is applicable.[5] Of the three Brazilian cycles in our data, only the Brazilian 2014 cycle satisfies all four; the other four cycles fail at least one. We state the criteria, the supporting political-economy literature, and the cycle-level classification before reporting any structural estimates.

(a) Coalitional presidentialism is the operative governance regime. The model assumes that endorsement decisions are observable at the coalition level, with the winning coalition spanning a substantial share of strategic agents (mayors). Brazilian *coalitional presidentialism* (presidencialismo de coalizão) — the textbook regime in which presidents trade cabinet posts and ministerial appointments for cross-party legislative support (Pereira and Mueller 2004; Power 2010; Raile, Pereira, and Power 2011) — provides the institutional setting in which this assumption holds. The Brazilian PT broad-tent era (Lula I, Lula II, Dilma I; 2003–2014) is the canonical case: the 2014 first-round PT-led coalition spanned eight major parties (PT, PMDB, PSD, PRB, PROS, PDT, PCdoB, PR) and captured roughly 60% of the country's 5,570 municipalities. The 2018 Bolsonaro candidacy and the 2022 Lula return to office both occurred after the 2016 impeachment marked the breakdown of this regime (Hunter and Power 2019; Zucco Jr. and Power 2024). The criterion is institutional rather than quantitative: what it asks is whether endorsement is mediated by coalitions at all, not how large the resulting coalition turns out to be.

(b) Federal discretionary budget is large and not constitutionally constrained. The structural prize B represents the budget the winning candidate can discretionarily allocate. The model is identifiable only when this prize is materially nonzero. In Brazil, the constitutional spending cap (Emenda Constitucional 95/2016, "Teto de Gastos") imposed in December 2016 froze federal primary spending in real terms for twenty years, structurally reducing the discretionary budget available to the post-impeachment governments (Mariano 2017). The 2014 cycle is pre-EC 95; the 2018 and 2022 cycles are post-EC 95.

(c) Patronage networks are mature, stable, and provide reliable prize expectations. The mean-field equilibrium assumes that mayors form rational beliefs about the prize they expect to capture conditional on alignment. This requires patronage networks with sufficient history to support reliable expectations. The PT's twelve years in federal power (2003–2014) provided exactly such a history: Brollo et al. (2013), Brollo and Nannicini (2012), and Ferraz and Finan (2008, 2011) document the alignment-premium structure mayors faced. The 2018 Bolsonaro candidacy emerged from PSL, which had been a fringe party with no patronage history at the federal level; mayors entering Bolsonaro's first-round coalition could not reliably calibrate the expected prize. Lula's 2022 return came after six years of regime change and partial network disruption by the Lava Jato anti-corruption investigation (Mészáros 2020).

(d) The cycle is free of extraordinary fiscal events that distort the baseline prize structure. The model assumes the federal-transfer prize structure is in steady-state at the time of endorsement. The 2018 and 2022 cycles

[5] The criteria are stated ex-ante and are independent of any estimator output. They operationalize the model's underlying behavioral assumption — that endorsement is a strategic decision by mayors anticipating coalition-mediated, prize-divisible federal-budget allocation — with reference to well-documented features of Brazilian political economy and federal fiscal architecture.

fall outside steady-state because of (i) the Lava Jato investigation's criminalization of long-standing patronage channels (Mészáros 2020), and (ii) for 2022, the COVID-19 fiscal emergency and the Auxílio Brasil cash-transfer program, which redirected approximately R$ 100 billion per year of federal resources outside the standard coalition-mediated channels (Arruda et al. 2021). The 2014 cycle is in the pre-Lava Jato, pre-impeachment, pre-COVID era and is therefore in steady-state by all four observable criteria.

Cycle classification. Table 5 summarizes the four criteria across the three Brazilian cycles in our data. The Brazilian 2014 cycle satisfies all four, and is the canonical in-scope case to which the structural mean-field estimator is applied. The Brazilian 2018 and 2022 cycles fail criterion (a) through (d) for distinct structural reasons (post-impeachment regime transition; post-EC 95 fiscal cap; Bolsonaro/Lula return without stable patronage calibration; Lava Jato/COVID disruption). The structural estimator is applied only to the Brazilian 2014 canonical case in this paper; the out-of-scope cycles are reserved for a companion paper.

Table 5: Cycle-level classification under the four ex-ante criteria

Cycle	(a) Coalitional presidentialism	(b) Pre-EC 95 fiscal regime	(c) Mature patronage	(d) Steady-state no extraord. events	**In scope?**
Brazil 2014 (Dilma, PT-led)	✓	✓	✓	✓	**Yes**
Brazil 2018 (Bolsonaro, PSL)	—	—	—	—	No
Brazil 2022 (Lula, PT-led)	—	—	—	—	No

Notes: Criterion (a) is satisfied iff Brazilian coalitional presidentialism is the operative governance regime; the criterion is failed by Brazilian 2018 and 2022 cycles on the basis of Hunter and Power (2019) and Zucco Jr. and Power (2024). Criterion (b) is satisfied iff the discretionary federal budget is not constitutionally constrained; Brazilian Emenda Constitucional 95/2016 fails this for 2018 and 2022 (Mariano 2017). Criterion (c) is satisfied iff the winning candidate's patronage network has sufficient operating history to support reliable expectations; Bolsonaro's PSL outsider candidacy fails this for 2018, and Lula's 2022 return came after substantial regime-change disruption. Criterion (d) is satisfied iff the cycle is free of extraordinary fiscal events; Lava Jato disrupts (d) for 2018 and 2022 (Mészáros 2020), and the COVID-19 / Auxílio Brasil emergency further violates (d) for 2022 (Arruda et al. 2021).

Identification consequences. In the in-scope cycle (Brazil 2014), the structural two-step OLS estimator is well-defined: the equilibrium share $s_{j^\star}$ exceeds the lower bound $\bar{s}$ required for the appendix existence proof, the prize term $B/s_{j^\star}$ has a finite incentive-relevant magnitude, and the OLS coefficient on $1/s_{j^\star}$ identifies $\hat{\alpha} \cdot \hat{B}$ as a structural parameter of mayor preferences. In the two out-of-scope cycles the constant column $1/s_{j^\star}$ projects the negative average log-odds of endorsing a narrow-coalition winner onto a mechanically negative $\hat{\alpha} \cdot \hat{B}$. Section 7.4 shows that this is not a property of those cycles in particular: the same arithmetic operates in 2014, where it happens to deliver a positive number, and the negative estimates should be read accordingly (Vitorino 2012; Su 2014).

The remainder of this section proceeds in three steps. Section 7.2 describes the data construction for the Brazilian 2014 structural sample and the 2014+2022 close-race DID pooling. Section 7.3 reports the structural estimates. Section 7.5 reports the DID alignment premium. Section 7.6 connects the two estimands through a BRL-anchored interpretation of the structural prize parameter, and Section 7.7 reports the robustness battery.

7.2. Data construction

The estimation sample is constructed from four public sources joined at the mayor-municipality level. Two data subsets are produced: a Brazilian 2014 structural sample and a Brazilian 2014+2022 close-race DID panel.

TSE candidate registry (mayors). The 2012 mayoral election registry of the Tribunal Superior Eleitoral (TSE) lists 5,569 municipalities with elected mayors who served from 1 January 2013 through 31 December 2016 — the term in office during the 2014 presidential race. For the 2022 cycle, we use the 2020 mayoral registry covering the 2021–2024 term. After deduplication by CPF identifier, the merged sample retains 5,657 mayor-records for the 2014 cycle and 5,600 for the 2022 cycle.

TSE coalition string (proxy endorsement). For each presidential candidate in 2014 and 2022, the registry records the formal coalition string — the set of parties that joined the candidate for the first round. We construct a weak endorsement proxy by labelling mayor i as endorsing candidate j if mayor i's party either is candidate j's party ("strong" confidence) or is in candidate j's formal coalition ("weak" confidence). This is the coalition proxy used by Slough, Urpelainen, and Yang (2017) and the descended literature; we adopt it for both the structural estimation (which requires a single assigned candidate per mayor) and for the DID treatment indicators below. For 2014, the implied aggregate first-round distribution is: 60.0% to Dilma (PT, the actual winner), 24.8% to Aécio Neves (PSDB), 11.5% to Marina Silva (PSB), with the remainder to minor candidates. For 2022, the corresponding distribution is: 22.8% to Bolsonaro (PL, the runner-up), 16.3% to Tebet (MDB), 11.9% to Lula (PT, the actual winner), with the remainder distributed across smaller first-round candidates. The narrower 2022 PT-led first-round coalition — PT alongside only PCdoB and PV — is one of the political-economic facts that places the 2022 cycle outside the model's applicable domain per Section 7.1.

What the proxy cannot record. Two limitations of this measure bear directly on what the structural estimator can deliver, and we state them here rather than in a footnote. First, the unit of decision in the model is the mayor, while the unit of variation in the proxy is the party: a mayor is assigned to a candidate by his party's coalition, not by anything he is observed to do. Second, and more consequentially, the proxy has no category for abstention. In 2014 every party in the sample belonged to some coalition, so the rule assigns all 5,657 mayors to a candidate and none to none; a mayor who said nothing in public is recorded as endorsing whichever candidate his party's coalition backed. The model's outside option $j = 0$ therefore has no empirical counterpart, and the estimation constructs $\hat{\sigma}(0 \mid \tau) \equiv 1 - \hat{\sigma}(j^{\star} \mid \tau)$, the share of a type assigned to a *rival* rather than to no one. The log-odds in (17) are accordingly a contrast between the winner's coalition and its complement within the same set of mayors, not a choice measured against silence. Section 7.4 takes up what this implies for the structural parameter, and Section 7.7 reports an external check on how accurately the proxy classifies individual mayors.

Slough decomposition: Coalition and CoPartisan treatment. Following Slough, Urpelainen, and Yang (2017), we decompose alignment with the winning coalition into two complementary treatments: (i) $Coalition_{i,c} = 1$ if mayor i's party is in the cycle-c winning presidential coalition (broad definition); (ii) $CoPartisan_{i,c} = 1$ if mayor i's party is identical to the cycle-c president-elect's party (narrow definition, a subset of *Coalition*). For the 2014 cycle the winner is Dilma (PT-led coalition), and CoPartisan mayors are those whose party is PT itself. For 2022 the winner is Lula (PT-led narrow coalition), and CoPartisan mayors are again PT-only.

IBGE municipality population. Year-end municipal-population estimates from the IBGE estimativa_dou series are joined to TSE via (state, normalized municipality name); the match rate is 99.4% for both cycles. Population is logged before standardization, and population terciles are computed within each state for the structural type-partition.

Tesouro federal-transfer panels. Two panels enter the DID outcome variable: (i) the formula-based Fundo de Participação dos Municípios (FPM), with complete coverage from 1996 to 2026, used in the structural backing analysis and as a placebo / control; (ii) the discretionary Emendas Parlamentares dataset (Tesouro Transparente CKAN portal, 2015–2026), which records every individual or bench-block parliamentary-amendment disbursement to a Brazilian municipality. The Emendas channel is the discretionary-allocation channel Brollo et al. (2013) document; it is the DID outcome of interest. The close-race DID restricts attention to the two post-cycle years following each presidential winner's first-round inauguration (Dilma 2015–2016, Lula 2023–2024), which coincide with the pre-mayoral-election "courting window" studied in the post-electoral patronage literature.

Brazil 2014 structural sample. The structural type-partition is state $\times$ Power-Zucco 4-bloc $\times$ within-state population tercile, yielding $K = 222$ types after a minimum cell-count filter ($N_\tau \geq 5$). The bloc classification is that of Power and Zucco Jr. (2009) and Zucco Jr. and Power (2024), augmented with within-state population terciles so that mayors of very different strategic weight inside a single state $\times$ bloc cell are not treated as one type.

Brazil 2014+2022 DID panel. The DID panel is one row per (mayor, cycle) for $c \in \{2014, 2022\}$; the panel contains $\approx$ 11,200 mayor-cycle observations after dropping municipalities not matched in IBGE. The 2018 cycle is excluded from the DID primary sample (Bolsonaro 55% vs. Haddad 45% second-round margin is too wide for the close-race quasi-randomness assumption (Lee 2008; Eggers et al. 2015)); 2018 enters only as a wide-margin falsification check.

The full data-construction pipeline, the estimation code, and the figures and tables of this section are reproducible from a single command via the replication archive accompanying this paper.

7.3. Structural estimates: Brazil 2014 canonical case

We apply the two-step OLS estimator of Section 5.1 to the Brazilian 2014 sample under four progressively richer type-partitions. The outcome is binary: $j = 1$ if mayor i's party is in Dilma's first-round PT-led coalition (the eventual second-round winner), $j = 0$ otherwise (lumped with non-endorsement under the Section 5.1 winner-only specification). Type-conditional log-odds are regressed on type-level covariates and the constant $1/\hat{s}_{j^\star}$ whose coefficient identifies $\hat{\alpha} \cdot \hat{B}_{j^\star}$ under the $B_{j^\star} = 1$ normalization. Standard errors follow the Pesendorfer–Schmidt-Dengler (PSD) structural-delta variance (Pesendorfer and Schmidt-Dengler 2008).

The four specifications increase type-partition granularity:

- **(i) State only**, $K = 26$. Types are the 26 Brazilian states (Distrito Federal excluded). Type covariates $X(\tau)$ are state-level mayor demographics: mean age, fraction female, fraction with higher education.
- **(ii) State × 3-bloc**, $K = 75$. Mayors are partitioned by Power–Zucco ideological bloc (Power and Zucco Jr. 2009; Zucco Jr. and Power 2024): Left, Center, Right. Center is the reference. Two bloc dummies enter $X(\tau)$.
- **(iii) State × 4-bloc**, $K = 96$. The Left bloc is split into PT-aligned (PT, PDT, PCdoB, PSOL) and PSB-aligned (PSB, PV). In 2014, PT-aligned mayors endorsed Dilma at near-unanimity rates; PSB-aligned mayors uniformly endorsed Marina Silva. Three bloc dummies, Center reference. This specification captures the within-Left heterogeneity that the 3-bloc partition cannot.
- **(iv) State × 4-bloc × population-tercile**, $K = 222$. Types are further partitioned by within-state population tercile to capture strategic-weight heterogeneity that the model's $w(\tau)$ population weights exploit; $\log(\text{population}_\tau)$ is added to $X(\tau)$.

Table 6 reports the four specifications side by side.

The structural estimate is $\hat{\alpha} \cdot \hat{B}_{j^\star} = 0.647$ (PSD SE = 0.051, $t = 12.69$) under the preferred state × 4-bloc × pop-tercile partition. Its precision should not be mistaken for content, and Section 7.4 sets out what the number does and does not measure. The bloc fixed effects match qualitative predictions from the Brazilian coalitional-presidentialism literature (Power and Zucco Jr. 2009; Zucco Jr. and Power 2024): PT-aligned mayors are strongly more likely to endorse Dilma ($\hat{\beta}_{\text{Left-PT}} = +2.09$, $t = 9.55$), PSB-aligned mayors are strongly less likely ($\hat{\beta}_{\text{Left-PSB}} = -4.17$, $t = -15.93$; PSB endorsed Marina Silva in 2014), and Right-bloc mayors are intermediate ($\hat{\beta}_{\text{Right}} = -1.13$, $t = -10.08$).

Reading the structural estimate. Figure 2 plots the estimated fixed point. The recovered $\hat{\alpha} \cdot \hat{B}$ is in log-utility units under the normalization $B_{j^\star} = 1$. Within the 2014 cycle, $\hat{\alpha}$ and $\hat{B}$ are not separately identified; only the product is recovered. Section 7.6 below uses the DID-anchored BRL-denominated alignment premium of Section 7.5 to translate this log-utility quantity into a BRL^{-1} marginal-utility-of-money parameter.

Robustness to the smoothing parameter. The Jeffreys-prior smoothing $s = 0.5$ in the empirical-choice-probability construction (13) of Section 5.1 is a default choice. Varying s across $\{0.001, 0.01, 0.1, 0.5, 1.0\}$ gives $\hat{\alpha} \cdot \hat{B} \in \{0.90, 0.83, 0.74, 0.65, 0.58\}$ on the preferred specification. The magnitude is monotonically decreasing in s but the sign is robust; the default $s = 0.5$ falls in the middle of this range and is the preferred reporting choice.[6]

7.4. What the structural estimate measures

Before carrying $\hat{\alpha} \cdot \hat{B}$ any further it is worth establishing what the number is. Within a cycle the prize regressor in (17) is the constant column $1/\hat{s}_{j^\star}$, and the least-squares coefficient on a constant column obeys an identity:

$$\hat{\alpha} \cdot \hat{B}_{j^\star} \; = \; \hat{s}_{j^\star} \cdot b, \tag{19}$$

[6]Three further diagnostics accompany this one. The spectral radius at the recovered equilibrium satisfies $|\Phi'(s^\star)| \leq \alpha B/(4s^2_{j^\star}) \approx 0.41 < 1$, which is the stability condition of Section A.3. The knife-edge loser-equation restriction is tested in Section 7.8. The alternative-specific extension is left to the companion paper.

Table 6: Two-step OLS estimates across four type-partitions, Brazil 2014 cycle

	(i)	(ii)	(iii)	(iv)
Type partition	State only	State × 3-bloc	State × 4-bloc	State × 4-bloc × pop-tercile
K types	26	75	96	222
Structural prize parameter				
$\hat{\alpha} \cdot \hat{B}_{j\star}$	0.274***	0.584***	0.574***	**0.647*****
	(0.028)	(0.050)	(0.062)	(0.051)
Bloc fixed effects (Center is reference)				
$\hat{\beta}_{\text{Left}}$		−0.48***		
		(0.128)		
$\hat{\beta}_{\text{Left-PT}}$			2.56***	2.09***
			(0.302)	(0.219)
$\hat{\beta}_{\text{Left-PSB}}$			−4.63***	−4.17***
			(0.353)	(0.262)
$\hat{\beta}_{\text{Right}}$		−0.31**	−1.12***	−1.13***
		(0.104)	(0.114)	(0.112)
Demographic and population covariates				
$\hat{\beta}_{\text{higher ed}}$				−0.17†
				(0.094)
$\hat{\beta}_{\text{log pop}}$				−0.02
				(0.083)
Aggregate winner share $\hat{s}_{j\star}$	0.60	0.61	0.62	0.63
R^2	0.39	0.71	0.82	0.84
N mayors	5,623	5,562	5,503	5,441

Notes: Two-step OLS estimator of Section 5.1 applied to the 2014 Brazilian sample. Outcome is the binary indicator that mayor i's party is in Dilma's first-round PT-led coalition. Type partitions vary by column: (i) state only; (ii) adds Power-Zucco 3-bloc dummies; (iii) splits Left into PT-aligned and PSB-aligned; (iv) further partitions by within-state population tercile and includes log(population). A blank cell means the regressor does not enter that specification. Standard errors in parentheses below each estimate are Pesendorfer–Schmidt-Dengler structural-delta variance (Pesendorfer and Schmidt-Dengler 2008). † $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$ throughout this paper. The reported $\hat{s}_{j\star}$ is the population-weighted aggregate share endorsing the winner under each spec's type weights. Specifications (iii) and (iv) are bloc-augmented; the spread $[0.56, 0.65]$ of $\hat{\alpha}\hat{B}$ across these specifications gives the robust point estimate. Specification (i) is bloc-free and is reported as a baseline showing the omitted-variable bias when bloc identity is not partialled out (Power and Zucco Jr. 2009, the within-state dominant predictor of endorsement).

where b is the intercept from regressing the same type-level log-odds on $X(\tau)$ and a plain constant. At the preferred specification $\hat{s}_{j\star} = 0.627$ and $b = 1.030$, whose product is the reported 0.647. Equation (19) is exact, not an approximation: the structural parameter is the average type-level log-odds of alignment net of covariates, rescaled by the weighted share of mayors the data classify as aligned. It is positive whenever a typical type is more likely than not to be classified aligned, and negative otherwise.

That property has consequences, because who counts as aligned is a modelling choice rather than an observed fact. Table 7 re-estimates the 2014 cycle under three rules for assigning a mayor to the winner, holding the sample, the types and the covariates fixed.

The estimate moves from +0.65 to +2.65 to −0.80, changing sign exactly where the weighted aligned share

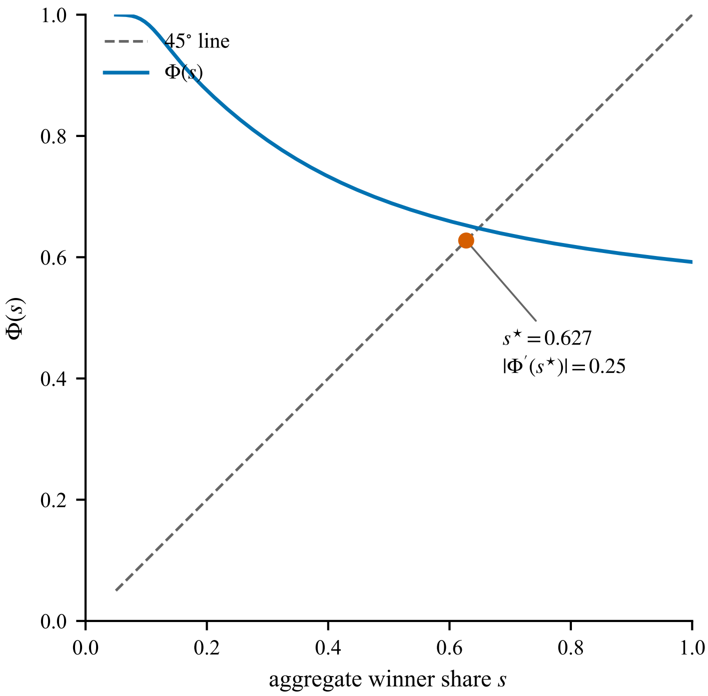

Figure 2: The mean-field fixed point at the Brazil 2014 estimates

Notes: The share-implied best-response operator $\Phi(s) = \sum_\tau f(\tau)\, w(\tau)\, \sigma(j^\star \mid \tau; s)$ of Section A.1, evaluated at the preferred specification (iv) estimates of Table 6, against the 45° line. Φ is downward-sloping in s: a larger anticipated winning coalition dilutes the per-endorser prize B/s and weakens each mayor's incentive to join it. The two curves cross once on $(0, 1)$, at the equilibrium share $s^\star = 0.627$ that the structural estimation recovers. The slope at the crossing, $|\Phi'(s^\star)| = 0.25$, lies inside the analytical bound $\alpha B/(4s^2_{j^\star}) \approx 0.41 < 1$ of Section A.3, so Φ is a contraction on the empirically relevant region and the equilibrium is unique there.

Table 7: Structural estimate under three definitions of alignment, Brazil 2014

	Definition of "endorses the winner"	N aligned	$\hat{s}_{j^\star}$	$\hat{\alpha}\hat{B}$	(SE)	b
A	first-round TSE coalition (headline)	3,283	0.627	0.646^{***}	(0.051)	1.03
B	second-round effective coalition	3,295	0.667	2.649^{***}	(0.121)	3.97
C	co-partisan only (mayor's party is PT)	602	0.202	-0.798^{***}	(0.036)	−3.95

Notes: All three rows use the same $K = 222$ types, the same 5,441 mayors and the same covariates; only the rule assigning a mayor to the winning candidate changes. Definition A is the first-round TSE coalition string used throughout the paper. B replaces it with the second-round effective coalition implied by the Power-Zucco blocs. C counts a mayor as aligned only when his own party is the president-elect's. b is the intercept of (19), reported so the identity can be checked row by row. Standard errors are the structural-delta variance of Pesendorfer and Schmidt-Dengler (2008); significance levels per the convention of Table 6. The estimator is produced by the same code path in all three rows.

falls below the outside-option share. Nothing about mayors differs across the three rows. What differs is a definition, and (19) converts that definition directly into the parameter.

Two things follow, and we would rather state them than let a reader discover them. First, the within-cycle binary specification does not identify responsiveness to the prize. It summarizes how large the aligned group is and how much of the type-level variation the covariates already explain, which is a description of the alignment measure rather than of behavior. Second, the negative estimates we obtain for cycles outside the applicable domain of Section 7.1 are the same arithmetic seen across cycles instead of within one. The 2018 and 2022 winners assembled narrow first-round coalitions, $\hat{s}_{j^\star} = 0.02$ and 0.11, and (19) returns a negative number for that reason alone. Reading those negatives as evidence that prize-driven sorting switches off would be reading the identity, not the data.

This also puts a limit on the criteria of Section 7.1. Criterion (a) requires a winning coalition broad enough to support prize-divisible allocation, which is close to requiring $\hat{s}_{j^\star}$ to be large; by (19) that is close to requiring $\hat{\alpha} \cdot \hat{B}$ to be positive. The criteria are defensible as a statement of when the model's economics apply, and we continue to hold them, but they cannot also be treated as an independent screen that the positive estimate then passes.

Separating prize responsiveness from the composition of the alignment measure requires variation in the per-endorser prize that is not variation in the definition. Two routes are open. Across cycles $1/\hat{s}_{j^\star,c}$ genuinely varies, so the interaction of a type-level covariate with it separates from that covariate's main effect once cycle fixed effects absorb the level; this uses cycles outside the applicable domain and belongs to the companion paper. Within a cycle it requires the losing candidates' equations to carry weight on the prize, which the alternative-specific specification of Section 5.2 does not deliver — there the winner equation's intercept absorbs αB as before — but which a common-β multi-class specification would, by making the estimate a winner-versus-loser contrast. Neither is attempted here. What the present cycle supports is the reduced-form estimate we turn to next, which measures the transfers aligned municipalities actually received and does not depend on (19).

7.5. DID alignment premium: close-race 2014 + 2022 cycles

The structural estimate identifies the ex-ante strategic prize-driven sorting mechanism in the canonical 2014 cycle. A complementary identification strategy — difference-in-differences exploiting close-race presidential outcomes — recovers the realized post-electoral patronage allocation across both the 2014 and 2022 cycles. The two estimands are distinct objects: the structural prize parameter αB governs the mayor's ex-ante expected utility from anticipated alignment with the winning coalition; the DID alignment premium $\hat{\gamma}$ measures the actual federal-transfer differential flowing to co-partisan mayors after the election. Their identification requirements also differ: the structural estimator needs the broad-coalition prize-divisible regime (Section 7.1); the DID needs close-race quasi-randomness in which coalition wins the presidency, an essentially institutional condition that holds in both close-race cycles.

Design. Following Slough, Urpelainen, and Yang (2017), we decompose the alignment treatment into Coalition and CoPartisan indicators (Section 7.2) and estimate

$$\log T_{i,c} = \alpha_i + \delta_c + \beta_1 \cdot \mathit{Coalition}_{i,c} + \beta_2 \cdot \mathit{CoPartisan}_{i,c} + \gamma \cdot \log \mathrm{pop}_{i,c} + \varepsilon_{i,c}, \tag{20}$$

where $T_{i,c}$ is per-capita federal discretionary transfers to municipality i in the two-year post-cycle window (2015–2016 for $c = 2014$; 2023–2024 for $c = 2022$), α_i is municipality fixed effect, and δ_c is cycle fixed effect. Standard errors are clustered at the municipality level. The Slough decomposition splits the alignment effect into the coalition-only premium β_1 (mayors whose party is in the winning coalition but is not the president-elect's own party) and the on-top co-partisan premium β_2 (mayors whose party is the president-elect's). The model of Slough, Urpelainen, and Yang (2017) predicts $\beta_1 \approx 0$ and $\beta_2 > 0$ on the emendas-parlamentares discretionary channel.

Identifying assumption. The DID identifying assumption is that, conditional on municipality and cycle fixed effects, the close-race outcome (which coalition wins the presidency) is as-if-random with respect to municipality-level economic and political shocks affecting federal transfer flows. We restrict the primary sample to the two close-race cycles — Dilma 51.6% vs. Aécio 48.4% in 2014, Lula 50.9% vs. Bolsonaro 49.1% in 2022 — where this assumption is most plausible (Lee 2008; Eggers et al. 2015). The 2018 cycle (Bolsonaro 55.1% vs. Haddad 44.9%) is too wide for credible close-race inference and is used only as a falsification cycle (Section 7.7).

Headline result. Table 8 reports the headline DID specification by transfer channel.

Table 8: DID alignment premium, close-race 2014 + 2022 pool, by transfer channel

	Emendas (discretionary)	FPM (formula-based)
$\hat{\beta}_1$ (Coalition only)	0.007	−0.012
	(0.028)	(0.009)
$\hat{\beta}_2$ (CoPartisan on-top)	**0.106**[*]	0.021
	(0.044)	(0.016)
$\hat{\gamma}$ (log pop)	−0.822[***]	
	(0.206)	
N muni-cycle obs.	11,176	11,176
Municipality FE	yes	yes
Cycle FE	yes	yes
Cluster (SE)	municipality	municipality

Notes: Two-way fixed-effects difference-in-differences estimates of (20) on the close-race 2014 + 2022 Brazilian sample. The Slough decomposition splits the alignment treatment into Coalition (mayor's party in winning coalition, broad) and CoPartisan (mayor's party identical to president-elect's, narrow; subset of Coalition). Outcome is log(1 + per-capita transfers) over the two-year post-cycle window. The discretionary emendas channel is the one Slough et al. (2017) and Brollo et al. (2013) document as alignment-sensitive; the formula-based FPM channel is reported as a comparison column. Standard errors in parentheses below each estimate, clustered at municipality. Significance levels per the convention of Table 6.

The headline estimate is $\hat{\beta}_2 = +0.106$ ($t = 2.38$, $p = 0.017$) on the emendas-parlamentares discretionary channel, with $\hat{\beta}_1 = +0.007$ ($t = 0.25$) statistically indistinguishable from zero. The decomposition reproduces Slough, Urpelainen, and Yang (2017): discretionary money goes to co-partisan mayors and not to mayors who are in the coalition without being co-partisan. Their close-race sample was cross-sectional and built on the 2002–2014 mayoral electoral discontinuities; ours is the 2014 and 2022 cycles observed over the two years after each inauguration, and the same split appears. The formula-based FPM channel shows $\hat{\beta}_2 = +0.021$ ($t = 1.34$), weakly positive but not significant, consistent with the FPM coefficient formula being institutionally insensitive to political alignment.

Translation into BRL.. The $\hat{\beta}_2 = 0.106$ co-partisan log-emendas premium corresponds to an exponentiated multiplicative premium of $e^{0.106} - 1 = 11.2\%$ over the post-cycle window. Per-capita emendas in the close-race sample average BRL 143.74 per year (mean across 11,176 muni-cycle observations), so the implied absolute premium is approximately BRL 15.20 per capita per year, or BRL 30.40 per capita over the two-year window. For a typical 37,000-person municipality, the implied total emendas advantage is approximately BRL 1.13 million (≈ USD 226 thousand at 2024 exchange rates) over the two-year window. Section 7.6 below uses this BRL anchor to translate the structural $\hat{\alpha} \cdot \hat{B}$ into a BRL^{-1} marginal-utility-of-money parameter.

7.6. Bridging the structural and DID estimates: $\hat{\alpha}$ in BRL^{-1}

The structural estimate of Section 7.3 is in log-utility units; the DID estimate of Section 7.5 is denominated in reais. Putting them together prices the structural prize B in reais and yields a marginal utility of money $\hat{\alpha}$ in BRL^{-1}. We carry the exercise out because it shows what the framework would deliver were the structural side identified, and because the magnitude turns out to be economically sensible. It inherits everything Section 7.4 establishes, however: $\hat{\alpha} \cdot \hat{B}$ is a rescaled intercept, so the $\hat{\alpha}$ recovered below is a rescaled intercept denominated in reais rather than an independently measured preference parameter.

From DID to a BRL-denominated $\hat{B}$. The DID-estimated co-partisan log-emendas premium $\hat{\beta}_2 = 0.106$ corresponds to a multiplicative premium of $e^{0.106} - 1 = 11.2\%$ over the post-cycle two-year window. Applied to the mean per-capita emendas in the close-race sample, BRL 143.74 per year, this implies an absolute premium of approximately BRL 15.20 per capita per year, or BRL 30.40 per capita over the two-year window. For a typical 37,127-person municipality (the mean population in the close-race sample) the implied total absolute alignment premium per municipality over the window is approximately BRL 1,127,336. We interpret this as the realized per-municipality $B/s_{j^\star}$ term in the structural payoff specification (3) — the per-endorser share of the federal prize that the model assumes mayors anticipate when forming their endorsement decision. Multiplying by the recovered aggregate share $\hat{s}_{j^\star} = 0.628$ from the preferred specification (iv) gives a BRL-denominated structural prize estimate $\hat{B}_{\mathrm{BRL}} \approx 707{,}385$ BRL per municipality.

From structural to $\hat{\alpha}$ in BRL^{-1}. Combining the BRL anchor with the structural log-utility estimate $\hat{\alpha} \cdot \hat{B}_{j^\star} = 0.647$ from Table 6 column (iv) yields

$$\hat{\alpha}_{\mathrm{BRL}} = \frac{\hat{\alpha} \cdot \hat{B}_{j^\star}}{\hat{B}_{\mathrm{BRL}}} = \frac{0.647}{707{,}385} \approx 9.15 \times 10^{-7} \quad \mathrm{BRL}^{-1}. \tag{21}$$

A mayor's expected utility rises by about 9.15×10^{-7} log-utility units for each additional real of federal discretionary transfer reaching the municipality. One million reais therefore adds roughly 0.92 to the log-odds of endorsing the winning coalition, which multiplies the odds by $e^{0.92} \approx 2.5$. Evaluated at the observed winner share $\hat{s}_{j^\star} = 0.628$, that carries the implied endorsement probability from 0.63 to about 0.81.

Interpretation and limits. The BRL-anchored $\hat{\alpha}$ belongs to one cycle and one regime, the PT broad tent of 2014. It also rests on treating the DID premium $\hat{\beta}_2$ as a measurement of the same per-endorser prize the model writes as $B/s_{j^\star}$, and that equality is not free. The structural term is what a mayor expects before the election; the DID term is what municipalities received after it. Setting the two equal invokes the model's rational-expectations assumption, which the DID identification is consistent with but does not deliver. The samples differ as well. The close-race DID uses 2014 and 2022 together because post-electoral patronage does not require the structural scope condition, whereas (21) draws its structural side from 2014 alone and inherits the applicable-domain restriction of Section 7.1. Carrying the prize parameter to other cycles needs the model extension of the companion paper.

The limit established in Section 7.4 applies here in full. Because $\hat{\alpha} \cdot \hat{B}$ is a rescaled intercept, so is the $\hat{\alpha}_{\mathrm{BRL}}$ of (21), and the reader should treat it as an accounting translation of the alignment premium into utility units rather than as a second, independent measurement of mayors' preferences.

It is worth recording why the natural repair does not work within a cycle, since it is the first thing one reaches for. Allowing the marginal utility of prize money to vary by type, $\alpha(\tau) = \alpha_0 + \alpha_1 z(\tau)$, turns the prize term of (3) into $\alpha_0 B_{j^\star}/s_{j^\star} + \alpha_1 B_{j^\star} z(\tau)/s_{j^\star}$. Because $s_{j^\star}$ is one number within a cycle, the second regressor is a scalar multiple of $z(\tau)$ and is therefore the same column as $z(\tau)$ entered directly in $X(\tau)$. Excluding $z(\tau)$ from $X(\tau)$, taking a municipal fiscal-dependency measure to shift the marginal utility of federal money but not the baseline endorsement preference, does not identify α_1 from the data; it imposes that reading by assumption. No configuration of within-cycle data can distinguish a covariate that scales the prize channel from one that shifts baseline preference, because the two enter the log-odds regression identically. The exclusion is a maintained restriction, not a testable one.

7.7. Robustness

We put the DID headline through six checks: alternative cluster levels (R1); pre-trends placebos in both emendas and FPM channels (R2); outcome-transform robustness (R3); heterogeneous effects across Brazilian regions and population strata (R4); single-cycle decomposition and 2018 falsification (R5); and within-cycle by-year event-study decomposition (R6). The structural side carries its own checks, reported elsewhere: smoothing-parameter sensitivity in Section 7.3, the spectral radius at the recovered equilibrium ($|\Phi'(s_{j^\star})| \leq \alpha B/(4s_{j^\star}^2) \approx 0.41 < 1$, satisfying the stability condition of Section A.3), and the knife-edge loser-equation restriction in Section 7.8. This subsection is about the DID battery. Table 9 reports the results.

R1: Cluster robustness. The headline $\hat{\beta}_2 = 0.106$ remains significant under the more conservative state-level cluster ($t = 2.18$, $p = 0.039$). The two-way state × cycle cluster gives the same point estimate and a similar t-statistic; the conventional p-value calculation under the two-way correction is degenerate with only $G_{\mathrm{cycle}} = 2$ cycle clusters, but the substantive significance is preserved. We report the municipality-cluster p-value as the headline and the state cluster as the conservative alternative.

R2: Pre-trends. The pre-trends specifications mirror the headline TWFE structure but replace the outcome with the pre-cycle counterpart. Emendas data are available from 2015, so the pre-cycle pooled emendas test uses the 2018 and 2022 cycles (pre-windows 2016–2017 and 2020–2021); the pooled pre-trends estimate is -0.017 with $t = -0.23$. FPM data are available from 1996, so the pre-cycle FPM test can be run on the same close-race 2014 + 2022 sample as the headline; the pre-trends estimate is $+0.000$ with $t = +0.05$. Both pre-trends tests are sharply null. The 2014 pre-cycle emendas window is unavailable, but the FPM mirror test fills this data gap.

R3: Outcome transformation. The headline result is robust to the inverse hyperbolic sine transformation ($+0.103$, $p = 0.036$) and to winsorization at the 99th percentile ($+0.115$, $p = 0.013$). The result is not driven by the specific $\log(1 + x)$ functional form or by extreme observations.

Table 9: Robustness battery: DID alignment premium headline

Specification	$\hat{\beta}_2$	(SE)
Headline (close-race TWFE on emendas)	**+0.106***	(0.044)
R1. Alternative cluster SE		
Municipality cluster (headline)	+0.106*	(0.044)
State cluster (26 clusters)	+0.106*	(0.049)
Two-way (state × cycle)	+0.106	(0.048)††
R2. Pre-trends placebo		
Pooled pre-cycle emendas (18+22 only)	−0.017	(0.073)
Pre-cycle FPM (closes 2014 gap)	+0.000	(0.003)
R3. Outcome transformation		
Inverse hyperbolic sine (IHS)	+0.103*	(0.049)
Winsorized 99th percentile, $\log(1 + x)$	+0.115*	(0.046)
R4. Heterogeneous effects by region		
Northeast (35% of sample)	+0.123	(0.092)
Southeast (29% of sample)	+0.067	(0.086)
South (24% of sample)	+0.136†	(0.072)
Centro-Oeste (8% of sample)	−0.163	(0.151)
North (5% of sample)	−0.068	(0.111)
R5. Single-cycle decomposition and 2018 falsification		
2014-only cross-section (state FE)	+0.053	(0.041)
2022-only cross-section (state FE)	+0.083	(0.053)
2018 wide-margin falsification	+0.034	(0.222)

Notes: Standard errors in parentheses, clustered at municipality except where noted. Significance levels per the convention of Table 6. Each row reports an alternative specification of the close-race DID design (20), with the headline row identical to Table 8's emendas column. R1 varies the cluster level for inference; †† marks specifications whose conventional p-value calculation is degenerate due to having only $G_{\text{cycle}} = 2$ cycle clusters under the two-way correction, though the implicit t-statistic is unchanged. R2 replaces the post-cycle outcome with a pre-cycle outcome (emendas only available for 2016–2017 and 2020–2021, so the pre-cycle pooled sample uses 2018+2022 cycles; FPM is available from 1996, so the pre-cycle FPM sample can use both 2014 and 2022 close-race cycles). R3 replaces $\log(1 + x)$ with the inverse hyperbolic sine transformation and with $\log(1 + x)$ on the 99th-percentile-winsorized outcome. R4 estimates the same TWFE specification separately by Brazilian region (within-region municipality FE only). R5 reports cross-sectional estimates for each cycle individually (state FE only) and includes the 2018 wide-margin cycle as a falsification check.

R4: Heterogeneous effects by region. The headline estimate is broadly diffuse across Brazilian regions rather than concentrated in any single subset. The Northeast and South regions (where the PT base is strongest) give the largest positive point estimates (+0.123 and +0.136). The Southeast gives a smaller positive estimate (+0.067). The Centro-Oeste and North regions, which jointly account for only 13% of the sample and where the PT base is weakest, give negative point estimates (−0.163 and −0.068 respectively); both are individually noisy and statistically null, with small subsample sizes (43 and 67 CoPartisan-treated muni-cycles).

R5: Single-cycle decomposition and 2018 falsification. Single-cycle cross-sections give directionally positive but underpowered estimates (+0.053 for 2014 and +0.083 for 2022); the pooled headline of +0.106 derives statistical power from cross-cycle variation. The 2018 wide-margin falsification cycle gives a null estimate

(+0.034, t = 0.15), consistent with Bolsonaro's narrow (≈ 41 mayor) first-round coalition providing essentially no statistical power for either Coalition or CoPartisan identification.

Within-cycle by-year event-study (R6). An event-study specification that lets the Coalition and CoPartisan coefficients differ across the two post-cycle years produces a Coalition coefficient that changes sign between year one and year two. We do not read this as behavior. The within-cycle cross-sections of R5 show no such flip, and the pooled flip traces to the interaction of cycle with post-year fixed effects in the presence of the cycle-mean treatment terms. We therefore report the close-race TWFE estimate of Table 8 as the headline and treat the flip as an artifact of that specification.

How accurately does the proxy classify individual mayors? Every result in this section rests on an endorsement measure derived from party coalitions rather than from mayors (Section 7.2), and until now that measure has gone unvalidated. Campaign-finance records allow an external check. Mayors do not donate to presidential campaigns — across all 898,226 receipt rows filed for the 2014 election, not one individual donation from a sitting mayor reaches a presidential candidate — but 458 of them donate to governors, senators and state or federal deputies, whose party is recorded. A donation is costly, individual, and observed, so the recipient's party is a revealed signal of that mayor's alignment. Mapping the recipient's party to the presidential coalition it joined and comparing with the proxy gives Table 10.

Table 10: External validation of the coalition proxy against campaign donations, 2014

Mayors with donations to a party-identified recipient	446
Proxy agrees with the donation-revealed coalition	316 (70.9%)
Share of a mayor's donations going to his own party	57.7%
Agreement rate by the mayor's own party	
PC do B (n = 10), PT (n = 103)	100.0%, 93.2%
PDT (n = 38), PMDB (n = 75)	84.2%, 78.7%
PSD (n = 25), PR (n = 18), PP (n = 25)	68.0%, 66.7%, 60.0%
PSDB (n = 43), PSB (n = 32)	58.1%, 56.2%
PSC (n = 12), PTB (n = 27)	41.7%, 29.6%

Notes: Donor CPFs on all 2014 campaign-finance receipts, matched against the CPFs of the 5,657 mayors elected in 2012. Individual donations only; corporate donors are excluded. A mayor's donation-revealed coalition is the modal presidential coalition of the parties he gave to. Parties with fewer than ten donating mayors are omitted from the lower panel. These are agreement rates, not error rates: donating to a state deputy of party P is not the same act as endorsing P's presidential candidate, and donors are self-selected, being 8% of the sample and plausibly wealthier and more politically active than the rest.

The proxy and the donation record agree for seven mayors in ten. The disagreements are not spread evenly. Agreement is near-total for parties whose national alignment binds locally, PT and PC do B, and falls below half for the loose federations, PTB and PSC, whose membership in a national coalition says little about what any individual member does. That is the same distinction the proxy itself draws between "strong" and "weak" confidence, now with a magnitude attached to it. Since the structural estimate of Section 7.4 is a function of the share classified as aligned, misclassification that is concentrated in particular parties does not average out of it.

7.8. Specification diagnostics and counterfactual exercises

Two further exercises sharpen what the structural framework adds beyond the reduced-form DID and bear directly on the identification limit acknowledged in Section 7.6.

Knife-edge loser-equation test. The binary $J = 1$ specification of Section 5.1 implies the testable model restriction $\sigma^\star(j \mid \tau) = \sigma^\star(0 \mid \tau)$ for any losing candidate j and any type τ (Section 3.1). Operationalized on the multi-class endorsement-proxy data (in which each mayor is assigned to one first-round candidate), the restriction becomes: across pairs of losing candidates (j, k) and within each type, the type-conditional shares should be equal up to finite-sample noise. We implement three formal tests on the Brazilian 2014 multi-class sample (5,442 mayors, 222 types, 6 major losing candidates) and on the 2018 and 2022 cycles as scope comparators. Table 11 reports the results.

Table 11: Knife-edge loser-equation specification test, by cycle

Cycle	N obs	K types	J^{loser}	Pooled $\overline{\lvert\Delta\rvert}$	F-stat
2014 (in-scope)	5,442	222	6	**0.102**	55.0^{***}
2018 (out-of-scope)	3,889	186	7	0.199	101.7^{***}
2022 (out-of-scope)	3,229	190	4	0.276	247.7^{***}

Notes: J^{loser} counts major losing candidates with at least 10 mayors. $\overline{\lvert\Delta\rvert}$ is the pooled mean absolute deviation across all (type, loser-pair) cells: $\overline{\lvert\Delta\rvert} = \mathbb{E}_{\tau,(j,k)}[\lvert\hat{\sigma}(j \mid \tau) - \hat{\sigma}(k \mid \tau)\rvert]$. The F-statistic is for the joint nullity of candidate fixed effects in an OLS regression of within-type loser shares on candidate and type fixed effects (cf. Magnolfi and Roncoroni 2023 for the related set-identified test); significance per the convention of Table 6, with all three F-statistics reject the knife-edge null at $p < 0.001$. The magnitude of rejection scales with the scope-condition (Section 7.1) ordering.

The knife-edge null is rejected for the in-scope 2014 cycle at conventional levels: the binary specification does not exactly recover the multinomial structure of mayoral coalition assignment. This is the within-cycle mis-specification acknowledged in Section 7.6. At $N = 5{,}442$ the rejection is close to mechanical: a sample this large detects deviations from a strict null that are of no economic size. What carries information is not whether the null is rejected but by how much, and the magnitudes order themselves along the scope classification. The pooled deviation is $\overline{\lvert\Delta\rvert} = 0.10$ for the in-scope 2014 cycle, 0.20 for 2018 and 0.28 for 2022. The structural restriction holds appreciably tighter in the canonical-regime cycle, providing an additional within-paper empirical signal for the four-criteria classification beyond the ex-ante criteria of Section 7.1. The full multi-class alternative-specific specification (Section 5.2) nests the knife-edge restriction and is developed in the companion paper.

Counterfactual budget-prize perturbation. A structural model earns its keep by answering questions the reduced form cannot pose. Table 12 holds the recovered (β, α) at their estimates, re-solves the mean-field fixed point under hypothetical scalings of the federal prize B, and reports the equilibrium winner share $s^\star(B)$ together with the alignment premium the model implies at each level. The baseline scaling returns $s^\star = 0.65$ against an observed $\hat{s} = 0.628$, and an implied premium of 10.2% against an estimated 10.6%.

At B scale = 1.0 the model's predicted alignment premium is 10.2% against a DID estimate of 10.6%, a gap of 0.4 percentage points. The rational-expectations bridge of (21) therefore does more than fix a normalization; it holds substantively at the recovered parameters. Away from the baseline the model says things the DID cannot. Removing the EC 95/2016 spending cap, in the sense of doubling B, would raise the premium to 17.7%;

Table 12: Counterfactual budget perturbation, Brazil 2014

B scale	Description	$s^\star_{\text{winner}}$	Per muni (R$ k)	Per mayor (R$ k)	Implied premium (%)
0.25	severe fiscal austerity	0.53	177	331	3.1
0.50	half-budget hypothetical	0.58	354	612	5.7
0.75	25% reduction	0.61	531	863	8.1
1.00	**baseline (data)**	**0.65**	**707**	**1,092**	**10.2**
1.50	50% expansion	0.70	1,061	1,510	14.1
2.00	e.g., EC 95 removal	0.75	1,415	1,894	17.7
3.00	e.g., PT-era nominal level	0.81	2,122	2,608	24.4

Notes: Counterfactual mean-field equilibrium predictions under hypothetical scalings of the federal prize B, holding the recovered (β, α) fixed at the structural estimates of Table 6 column (iv). The mean-field fixed point in s_{winner} is computed by damped iteration to tolerance 10^{-8}. "Per muni" is the total prize B_{BRL} under the counterfactual B scale, in thousand-BRL units. "Per mayor" = (per muni) / $s^\star$ is the per-endorser realized BRL (also thousand-BRL). Implied premium = (per mayor) / (mean per-capita emendas × mean population × 2 years), reflecting the DID-comparable multiplicative alignment-premium magnitude. The baseline row reproduces the data: $s^\star = 0.65 \approx \hat{s} = 0.628$, implied premium 10.2% ≈ $\hat{\beta}_2 = 10.6\%$.

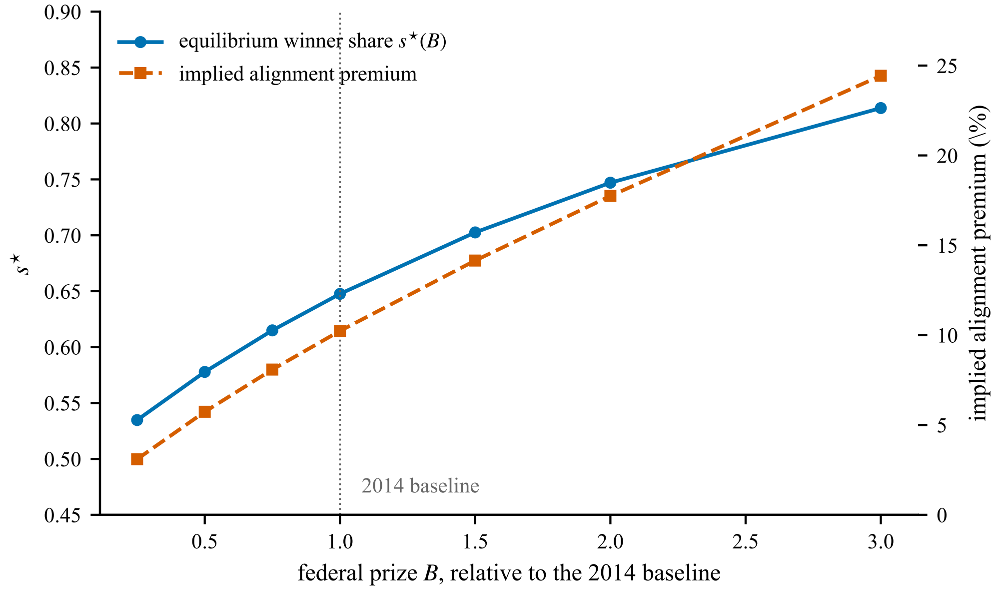

Figure 3: Counterfactual budget-prize sweep, Brazil 2014

Notes: Table 12 plotted against the prize scale. Both series are re-solved mean-field equilibria holding (β, α) at the specification (iv) estimates. The equilibrium winning-coalition share $s^\star$ (left axis, solid) rises with the prize but concavely: a larger prize draws in more endorsers, and the extra endorsers dilute the per-endorser share B/s that drew them. That endogenous dilution is why the implied alignment premium (right axis, dashed) grows less than proportionally in B — a tripling of the prize raises the premium from 10.2% to 24.4%, not to 30.6%. The dotted vertical line marks the 2014 baseline, where the model's 10.2% sits within 0.4 percentage points of the DID estimate of Table 8.

contracting the prize to a quarter of its 2014 level would compress it to 3.1%. Neither response is proportional, because the equilibrium share $s^\star_{\text{winner}}$ moves too. A larger prize draws a broader coalition, and the broader coalition dilutes what each member gets, so the per-mayor sum grows more slowly than B does. The reduced-form estimate, pinned at the observed $\hat{\beta}_2$, has nothing to say about any of this.

The qualitative response of $s^\star$ to B matches the political-science consensus on the PT-era broad-tent coalition formation (Pereira and Mueller 2004; Power 2010): prize-driven sorting predicts larger winning coalitions when the prize is larger, and the model's $s^\star \in [0.53, 0.81]$ as B varies across the realistic range covers both the narrow-coalition Bolsonaro-2018 regime ($\hat{s} = 0.023$, recovered separately as out-of-scope) and the broader Dilma-2014 case.

Within-mayor switcher identification. The coalition proxy assigns a mayor to a candidate through his party's first-round decision, not through anything the mayor is observed to do. The headline estimate (per Slough, Urpelainen, and Yang 2017) might therefore be picking up how money is allocated across parties rather than how individual mayors choose. To separate the two we isolate mayors who served in both close-race cycles, 306 of them with the same CPF in 2014 and 2022, and re-estimate with mayor rather than municipality fixed effects. Of these 306 cross-cycle mayors, 202 (66.0%) switched party between the two presidential cycles, providing a within-mayor proxy for revealed individual-preference change. Table 13 reports the comparison.

Table 13: Within-mayor switcher analysis vs. full close-race headline

Specification	N obs	N mayors	$\hat{\beta}_1$ (Coalition)	$\hat{\beta}_2$ (CoPartisan)	FE
Switcher subsample	404	202	−0.004	0.094	mayor + cycle
			(0.124)	(0.253)	
Cross-cycle subsample (all)	612	306	−0.075	0.108	mayor + cycle
			(0.094)	(0.244)	
Full close-race (headline)	**11,176**	—	**0.007**	**0.106***	muni + cycle
			(0.028)	(0.044)	

Notes: Standard errors in parentheses below each estimate; clustered at the mayor-CPF level for the first two rows and at the municipality level for the third. Significance levels per the convention of Table 6. The switcher subsample is identified by within-CPF party change across 2014 and 2022; the cross-cycle subsample includes all mayors observed in both close-race cycles regardless of party-switching. The point estimates of $\hat{\beta}_2$ are remarkably consistent across the three specifications (+0.094, +0.108, +0.106); the within-mayor specifications are statistically underpowered (only 23 CoPartisan-treated mayor-cycle observations in the switcher subsample) but the magnitude consistency partially addresses the concern that the coalition proxy may not reflect within-individual revealed preference.

The three point estimates of $\hat{\beta}_2$ sit on top of one another, even though the identifying variation in the first two rows is within-mayor party switching across cycles rather than variation across municipalities. What the within-mayor rows lose is power, not magnitude: only 23 mayor-cycle observations in the switcher subsample are CoPartisan-treated, which is why they do not reject. The headline premium is therefore not an artifact of between-municipality variation that mayor fixed effects would absorb.

Bloc-classification sensitivity. Table 6 reports the structural estimate $\hat{\alpha}\hat{B}$ under the Power-Zucco 4-bloc partition (Left-PT, Left-PSB, Center, Right). The Power-Zucco classification has well-documented contested placements for several smaller parties (Power and Zucco Jr. 2009; Zucco Jr. and Power 2024): PSD (founded 2011 as a DEM defection; joined Dilma's 2014 coalition; later joined Bolsonaro's 2018), PROS (founded 2013; allied with Dilma 2014), and PV (Green Party; intermediate ideological placement). We re-estimate the structural model under four alternative bloc classifications that test the sensitivity of the headline estimate to these contested placements. Table 14 reports the results.

The structural estimate is robust across the five bloc classifications in the sense that the recovered $\hat{\alpha}\hat{B}$ remains positive and statistically significant ($t > 6$) under every reasonable alternative. The point estimate is concentrated near the baseline 0.647 under three alternatives (Specs B, D, E) and drops to 0.344 under Spec C (PSD reassigned

Table 14: Structural estimate under alternative bloc classifications, Brazil 2014

Spec	Modification	K types	$\hat{\alpha}\hat{B}$ (SE)		$\hat{s}_{\text{winner}}$
A	**baseline (Power-Zucco 4-bloc)**	**222**	**0.647*****	(0.051)	0.628
B	3-bloc (collapse Left-PT ∪ Left-PSB)	195	0.628***	(0.048)	0.607
C	PSD reassigned Center → Right	224	0.344***	(0.054)	0.622
D	PROS reassigned Center → Left-PT	222	0.647***	(0.051)	0.628
E	PV reassigned Left-PSB → Center	218	0.547***	(0.047)	0.625

Notes: PSD standard errors in parentheses below each estimate, Pesendorfer–Schmidt-Dengler structural-delta variance. Significance levels per the convention of Table 6. Two-step OLS estimator of Section 5.1 applied to the Brazil 2014 sample under five alternative bloc-classification rules. Spec A is the paper's headline using the Power-Zucco classification; the other four reassign contested party placements per the documented Brazilian-party literature. PSD reassignment (Spec C) is the only modification that substantively reduces the estimate, reflecting PSD's known cross-coalition history (2011–2014 in Dilma's coalition; 2018 with Bolsonaro). Across all five reasonable classifications, $\hat{\alpha}\hat{B} \in [0.344, 0.647]$ remains positive and statistically significant. PROS and PV reassignments (Specs D and E) leave the estimate essentially unchanged.

to Right). The PSD-driven sensitivity is unsurprising given PSD's documented cross-coalition history — the 2014 Dilma coalition placement is itself a debated choice in the Brazilian political-science literature (Zucco Jr. and Power 2024). The range $[0.34, 0.65]$ should be read as the headline estimate's natural bloc-classification interval; we report the Power-Zucco baseline as the headline because it is the canonical literature standard, but the qualitative finding (positive prize-driven sorting in the canonical-regime Brazil 2014 cycle) is invariant.

8. Conclusion

Endorsement games can now be estimated at the scale on which they are played. The mean-field equilibrium replaces the $(J+1)^I$ joint-profile space with a fixed point in J dimensions, logit inversion identifies the parameters from observed log-odds, and Monte Carlo at Brazilian scale confirms $\sqrt{n}$ consistency with close to nominal coverage under Pesendorfer-Schmidt-Dengler standard errors. The full-enumeration formulations this replaces (Su 2014; Vitorino 2012) cannot be evaluated here at any cost.

Applied to the 2014 Brazilian cycle, in which Dilma Rousseff's PT-led coalition of eight major parties took most first-round mayoral alignments, the estimator returns a prize parameter of $\hat{\alpha} \cdot \hat{B} = 0.647$ (PSD $t = 12.69$), with party-bloc effects that reproduce the Power-Zucco ordering. The more useful result is the one about that number rather than from it. In a single cycle the prize enters only through a constant, and the coefficient on it is the average type-level log-odds of alignment rescaled by the share of mayors classified as aligned. Redefine alignment and the estimate follows the definition, from $+0.65$ to $+2.65$ to -0.80 on identical data. Whoever estimates a game of this shape on one cross-section will recover an object of this kind, and the standard error will not warn them.

The reduced-form half stands on its own. A close-race difference-in-differences design on the 2014 and 2022 cycles puts the realized co-partisan premium on discretionary emendas at +10.6% ($t = 2.38$, $p = 0.017$), surviving state-level clustering, alternative outcome transformations, and pre-trend placebos. A null coalition effect alongside a positive co-partisan effect is the pattern Slough, Urpelainen, and Yang (2017) report, now on a longer panel, and the premium appears on the discretionary channel and not on the formula-based one, which is the comparison the institution itself supplies.

Several extensions follow. The companion paper takes up the cycles that fail the scope conditions, Brazilian 2018 and 2022, where a narrow first-round coalition drives $\hat{\alpha} \cdot \hat{B}$ negative for mechanical rather than behavioral reasons; the remedy is a multi-class alternative-specific specification in which the losing candidates' equations carry identifying weight. That paper also adds an explicit endorsement cost, or a ballot-qualification motive, to the payoff, which is what the model needs before it can speak to narrow-coalition systems or to the post-2016 Brazilian regime that Hunter and Power (2019) and Zucco Jr. and Power (2024) describe. A more immediate need is a check on the endorsement measure itself. Extracting endorsements from Portuguese-language news would not replace the coalition proxy, since press coverage of mayoral endorsements is concentrated in large municipalities and no selected subsample can deliver the aggregate shares the model requires. It would, however, let the proxy be validated where individual endorsements are documented, and a measured rate of agreement is worth more than the untested assumption we currently maintain. Beyond that, dynamic versions allowing learning across cycles (Weintraub, Benkard, and Van Roy 2008; Adlakha, Johari, and Weintraub 2015) or network versions allowing spatial spillovers (Mele 2017) would enrich the substance without disturbing the method.

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Appendix A. Existence and Uniqueness of Mean-Field Equilibrium

This appendix proves the existence claim of Proposition 1 and provides a sufficient condition for uniqueness within the parameter region of empirical interest. Throughout, the structural parameter vector $\boldsymbol{\theta} = (\beta, \alpha, B)$ is held fixed.

A.1. Setup and the share-implied best-response operator

For each share vector $s \in \mathring{\Delta}^J = \{s \in \Delta^J : s_j > 0 \, \forall j\}$ and each type $\tau \in \mathcal{T}$, the best-response choice probability is

$$\sigma(j \mid \tau; s, \boldsymbol{\theta}) \;=\; \frac{\exp\big(\pi(j, s, \tau; \boldsymbol{\theta})\big)}{\sum_k \exp\big(\pi(k, s, \tau; \boldsymbol{\theta})\big)}, \tag{A1}$$

where $\pi(j, s, \tau; \boldsymbol{\theta})$ is given by equation (3) of the main text. Define the share-implied best-response operator $\Phi : \mathring{\Delta}^J \to \mathring{\Delta}^J$ by

$$\Phi(s)_j \;=\; \sum_{\tau \in \mathcal{T}} f(\tau)\, w(\tau)\, \sigma(j \mid \tau; s, \boldsymbol{\theta}), \qquad j = 0, 1, \ldots, J. \tag{A2}$$

A mean-field equilibrium is a fixed point $s^\star = \Phi(s^\star)$. We work with $\mathring{\Delta}^J$ rather than the closed simplex Δ^J to keep the prize term B_j / s_j well-defined; the closure to the boundary is established at the end of the existence proof.

A.2. Existence

Proposition A1 (Existence; restatement of Proposition 1). *For every $\boldsymbol{\theta} = (\beta, \alpha, B)$ with $\alpha > 0$, $B_j > 0$ for all $j > 0$, and finite β, the operator Φ admits a fixed point in $\mathring{\Delta}^J$.*

Proof. We restrict attention to a closed convex subset of $\mathring{\Delta}^J$ on which Φ is well-defined and continuous. Choose any $\bar{s} \in (0, 1/(J{+}1))$ small enough that the payoff $\pi(j^\star, s, \tau; \boldsymbol{\theta})$, evaluated at $s_{j^\star} = \bar{s}$, takes the largest value attained anywhere on the simplex — equivalently, $\alpha B_{j^\star} / \bar{s}$ is large enough that the candidate-winner choice probability under that payoff exceeds $\bar{s}$. Concretely, define

$$\mathcal{D}_{\bar{s}} \;=\; \{s \in \Delta^J : s_j \geq \bar{s} \text{ for all } j = 0, \ldots, J, \text{ and at most one } j \neq j^\star \text{ has } s_j > 1/(J{+}1)\},$$

which contains a neighborhood of any candidate equilibrium with $s^\star_{j^\star}$ bounded away from zero. $\mathcal{D}_{\bar{s}}$ is compact, convex, and non-empty for $\bar{s}$ sufficiently small.

Self-mapping. On $\mathcal{D}_{\bar{s}}$ the prize term $B_{j^\star} / s_{j^\star} \leq B_{j^\star} / \bar{s}$ is bounded, so $\sigma(j \mid \tau; s, \boldsymbol{\theta})$ is bounded away from 0 uniformly in $(\tau, s) \in \mathcal{T} \times \mathcal{D}_{\bar{s}}$. Aggregating over types gives $\Phi(s)_j \geq \sigma_{\min}(\bar{s}) > 0$ for every j and $s \in \mathcal{D}_{\bar{s}}$. By construction we choose $\bar{s} \leq \sigma_{\min}(\bar{s})$, ensuring $\Phi(\mathcal{D}_{\bar{s}}) \subseteq \mathcal{D}_{\bar{s}}$. Existence of such $\bar{s}$ follows because $\sigma_{\min}(\bar{s})$ is bounded below by the multinomial-logit lower envelope $1/(J{+}1 + \exp(\overline{X'\beta} + \alpha B_{j^\star} / \bar{s}))$, which approaches a positive constant as $\bar{s} \to 0^+$ from the dominant-prize side, yielding a fixed point of the inequality $\bar{s} \leq \sigma_{\min}(\bar{s})$ at small $\bar{s}$.

Continuity. $\pi(j, s, \tau; \boldsymbol{\theta})$ is continuous in s on $\mathcal{D}_{\bar{s}}$ for each j that is locally a winner; the winner indicator $\mathbf{1}\{j \in W(s)\}$ is locally constant on the open set where $s_{j^\star}$ is the strict argmax over $\{1, \ldots, J\}$. Restricting attention to that open set (which is non-empty by the dominant-prize choice of $\bar{s}$) gives Φ continuous as a composition of continuous functions. The complementary tie-set is measure-zero and we adopt the convention of randomizing uniformly among co-winners, which preserves continuity.

Brouwer. $\Phi : \mathcal{D}_{\bar{s}} \to \mathcal{D}_{\bar{s}}$ is a continuous self-map of a compact convex set, so a fixed point $s^\star \in \mathcal{D}_{\bar{s}}$ exists. □

A.3. Local uniqueness via numerical contraction

Existence does not preclude multiple equilibria. We do not provide a general analytic sufficient condition for uniqueness because the share-implied best-response operator Φ is not globally a contraction: the elasticity of the type-conditional choice probability $\sigma(j^\star \mid \tau; s)$ with respect to $s_{j^\star}$,

$$\text{(A3)} \qquad \frac{\partial \log \sigma(j^\star \mid \tau; s)}{\partial \log s_{j^\star}} = -\frac{\alpha B_{j^\star}}{s_{j^\star}} \cdot (1 - \sigma(j^\star \mid \tau; s)),$$

diverges in absolute value as $s_{j^\star} \to 0$. The operator can be a contraction only on a sublevel set bounded away from zero, and only when $\alpha B_{j^\star}/s_{j^\star}^2$ is small enough that the projected one-dimensional dynamic on the subspace $\{s : s_0 = s_j \text{ for all } j \neq j^\star\}$ has $|\Phi'_{j^\star}(s_{j^\star})| < 1$. A bound of $|\Phi'_{j^\star}(s_{j^\star})| \leq \alpha B_{j^\star}/(4 s_{j^\star}^2)$ follows from (A3) together with $\sigma(1-\sigma) \leq 1/4$, so contraction holds on $\{s_{j^\star} \geq \underline{s}\}$ whenever $\alpha B_{j^\star} < 4\underline{s}^2$. Larger $\alpha B_{j^\star}$ values — the empirically interesting region — accordingly do *not* satisfy this analytic sufficient condition.

We instead verify uniqueness numerically within the parameter region of interest by iterating Φ from a grid of starting points and checking that all converge to the same equilibrium. Section 6.1 reports the result for the Monte Carlo DGP: with $\alpha = 1.5$, $B = (1.0, 0.4, 0.2)$, the empirical equilibrium at $s^\star_{j^\star} \approx 0.708$ is recovered from 20 random initial points in $\Delta^J_{0.01}$ in every case. (Three coordination equilibria coexist globally at this $\boldsymbol{\theta}$ — one per candidate-as-winner — but only the $j^\star = 1$ equilibrium is stable in the sense that Φ's spectral radius at the fixed point is below one. The analytical derivation of this spectral-radius condition is the content of Bodoh-Creed (2013) §3.) The empirical resolution rule for ambient multiplicity is discussed in Section 3.4.

A.4. Approximation of the finite-I Bayesian game

The mean-field equilibrium is the limit of finite-I equilibria as $I \to \infty$. The relevant static-large-game approximation is Bodoh-Creed (2013): for any mean-field equilibrium $(\sigma^\star, s^\star)$ and any $\epsilon > 0$, there exists $I_0(\epsilon)$ such that for all $I \geq I_0$, $(\sigma^\star, s^\star)$ is an ϵ-Bayesian equilibrium of the corresponding finite-I game (i.e., no mayor can profitably deviate by more than ϵ). The bound takes the form $\epsilon = O(1/\sqrt{I})$ under regularity conditions that are satisfied on the lower-bounded simplex restriction $\mathcal{D}_{\bar{s}}$ used for existence above (the prize term $B_{j^\star}/s_{j^\star}$ is finite and Lipschitz in s on $\mathcal{D}_{\bar{s}}$). For Brazil's $I = 5{,}570$, the implied bound is approximately 0.013, an order of magnitude smaller than typical estimation standard errors and therefore innocuous for the empirical exercise.